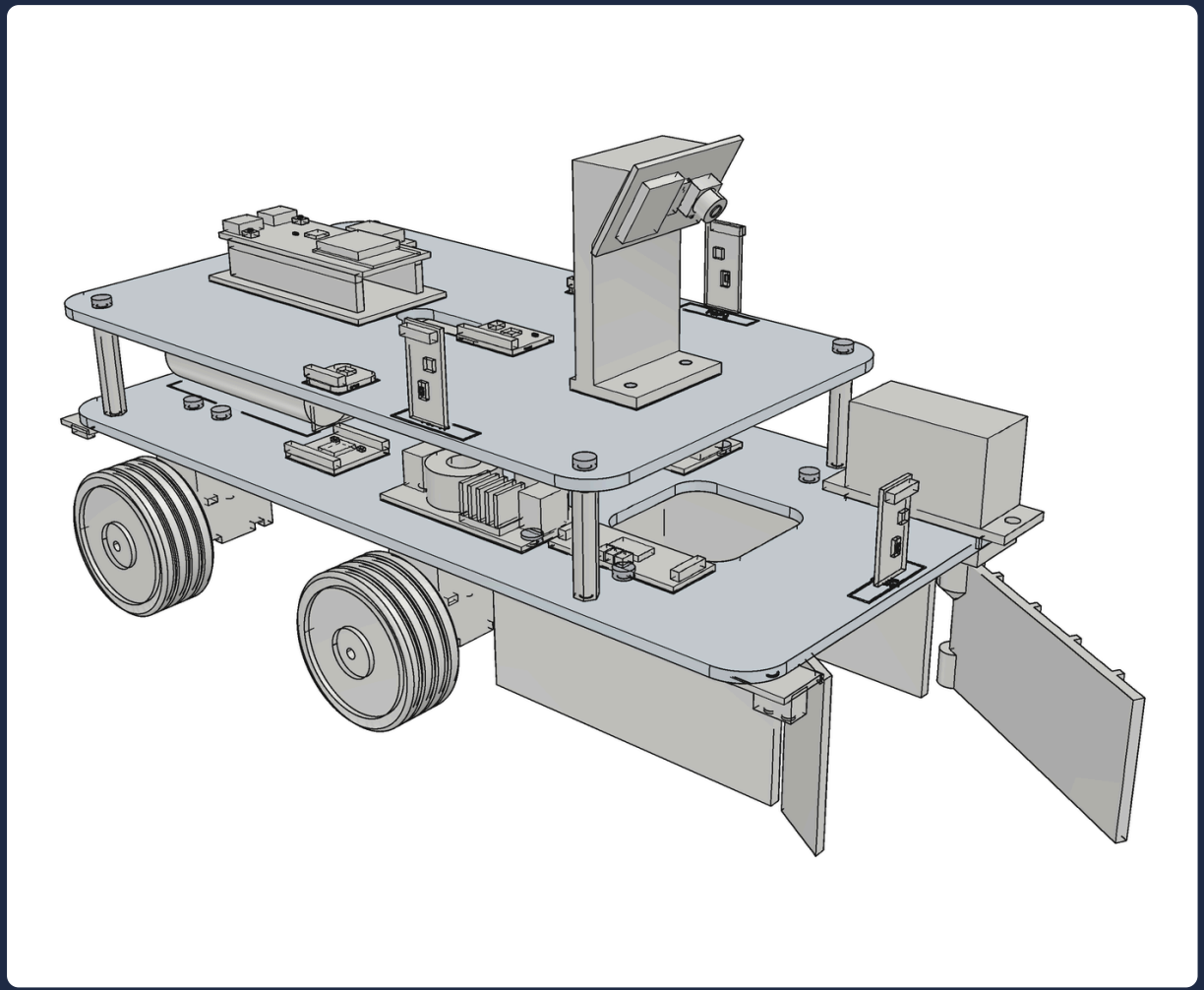
A.E.G.I.S. MK2

# Assembly guide

27 steps, one action each, from bare deck to finished rover



## BEFORE YOU START

**Tools:** Phillips screwdriver, side cutters, wire strippers, soldering iron, multimeter, hairdryer.

**Made first** (Fabrication guide): both laser-cut decks, all printed parts, the soldered ESP32-S3 carrier board, headers soldered on every module (Wiring guide W1-W3).

**Check** every board against its engraved outline before sticking it down.

**Battery** stays unplugged until step 27.

## HOW TO READ A STEP

added in this step  already fitted  screws, nuts

Orange arrows show where a part moves. RESULT shows the same view once the step is done. Numbered balloons match the YOU NEED list. An orange circle is a close-up. "x 4" next to the title: do it four times.

**Wiring** with AEGIS_Wiring_Guide.pdf at steps 16 (W4-W9) and 24 (W10-W15).

## Fasteners: everything comes from the M3 kit (item #15)

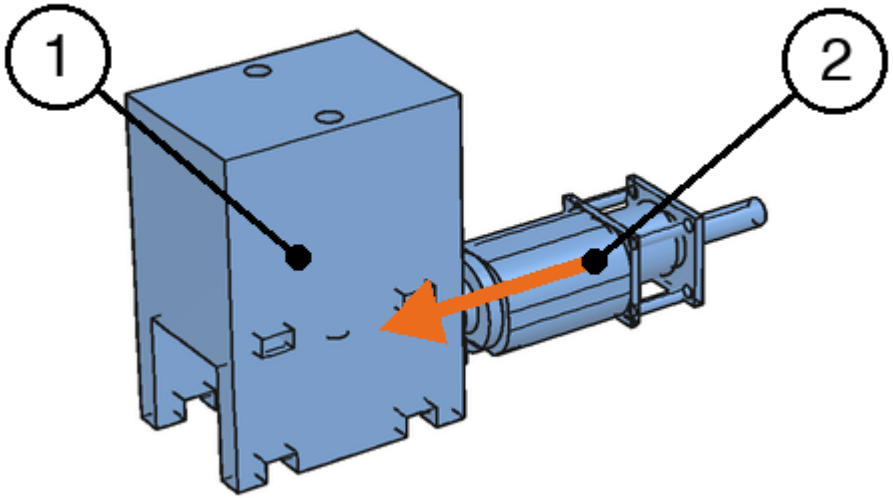
| Where | What | Qty | Step |
|---|---|---|---|
| Motor mounts to the lower deck | M3 x 6 screw, self-tapping into the PLA bridge | 8 | 3 |
| Intake shroud flanges | M3 x 6 screw, down through the deck into the 2.5 mm pilot | 4 | 6 |
| Pillars, lower end | 20 mm male-female pillar, stud through the deck + M3 nut | 4 + 4 | 17 |
| Pillars, upper part | 10 mm male-female pillar on top of the 20 mm one | 4 | 17 |
| Upper deck to the pillars | M3 x 6 screw | 4 | 18 |
| Camera mast | M3 x 6 screw, up through the deck into the 2.5 mm pilot | 2 | 25 |
| Servo ears, battery, motors | Cable ties, 100 mm (item #36) | 2 + 2 + 4 | 2, 7, 10 |
| All boards, brackets, camera | 3M foam tape (item #38) | about 1.5 m | 5, 9-15, 20-23, 26 |

**From the kit:** 18 of the 20 M3 x 6 screws (2 spare), 4 of the 20 nuts, 4 of the 10 pillars of each of the two male-female lengths. No other screws are needed. Self-tapping screws: turn them in slowly the first time and stop as soon as the head is snug, or the PLA thread strips.

**Printed parts** (plate 1, black PLA+): intake shroud, intake door, camera mast, four motor mounts, three L-brackets for the distance sensors. Plate 2 (orange PLA) holds the demo props: token and two decoys.

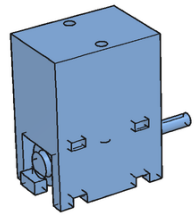
# 1 Motor into its printed mount

**x 4**



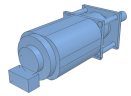

1. Slide the motor into the mount, gearbox first, until the gearbox is under the bridge.
2. The encoder cable leaves through the open back of the mount.
3. Make all four. Two face left and two face right: shafts point outward.

**RESULT**



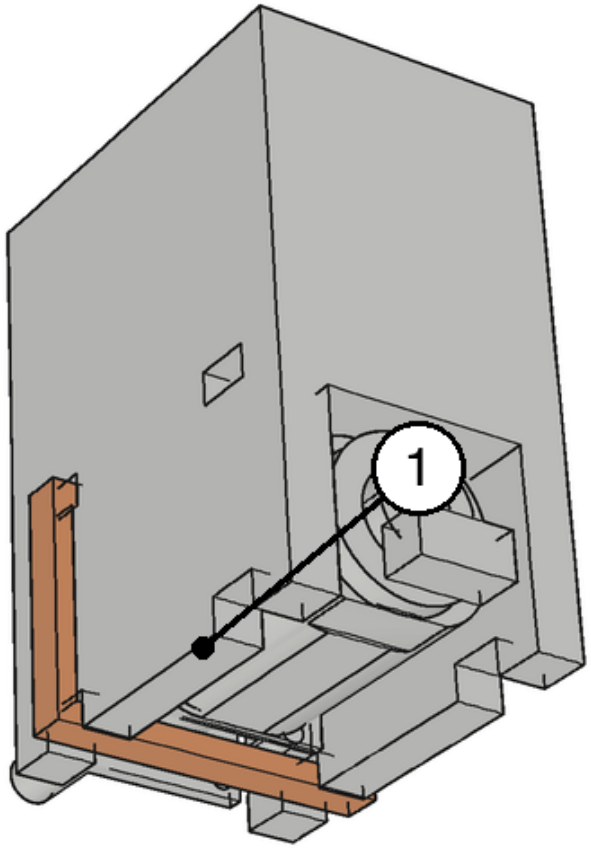
**YOU NEED**

1  Motor mount (printed) **x 4**
PLA, plate 1

2  N20 gearmotor + encoder **x 4**
item #5

# 2 Lock the motor with a cable tie

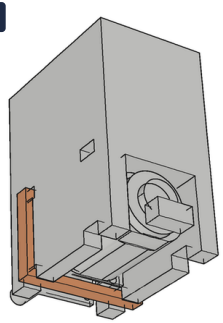
x 4




1. Push a cable tie through one channel in the bridge, down the outside of both cheeks,
2. under the motor through the two notches, and back into its head.
3. Pull snug and cut the tail. Use the second channel too if the motor still slides.

**Important.** Tight enough that the motor cannot slide, not so tight that it crushes the encoder board.

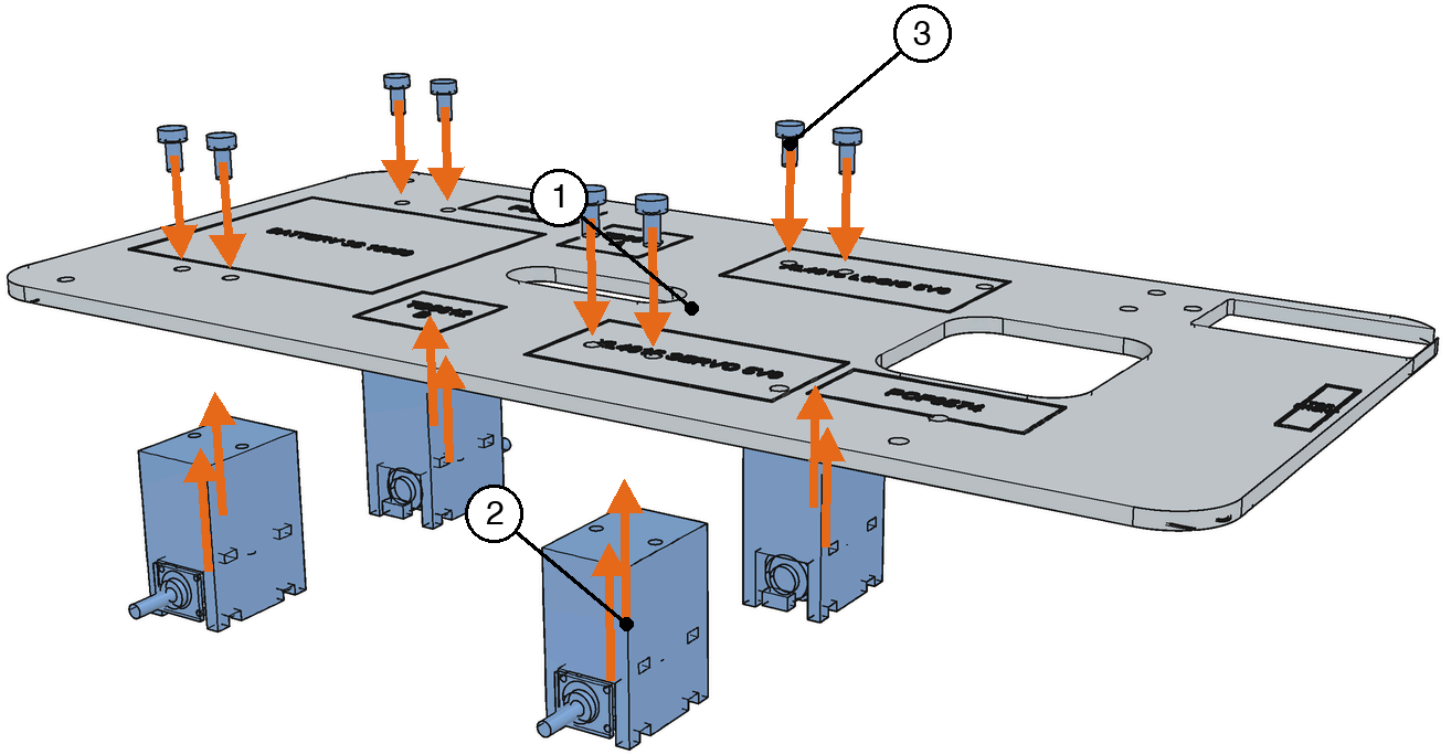
RESULT



YOU NEED

| | | |
|---|---|---|
| 1 |  | Cable tie, 100 mm x 4<br>item #36 |

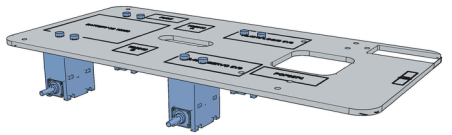
# 3 Mounts under the lower deck



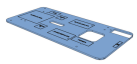

1. Peel the film off the deck and lay it engraved side up.
2. Hold each mount under its two holes, shaft outward, and drive an M3 x 6 down into each hole.
3. The screw cuts its own thread in the PLA: stop as soon as the head is snug.

**Important.** Keep each mount square to the deck edge, or the rover will drive crooked.

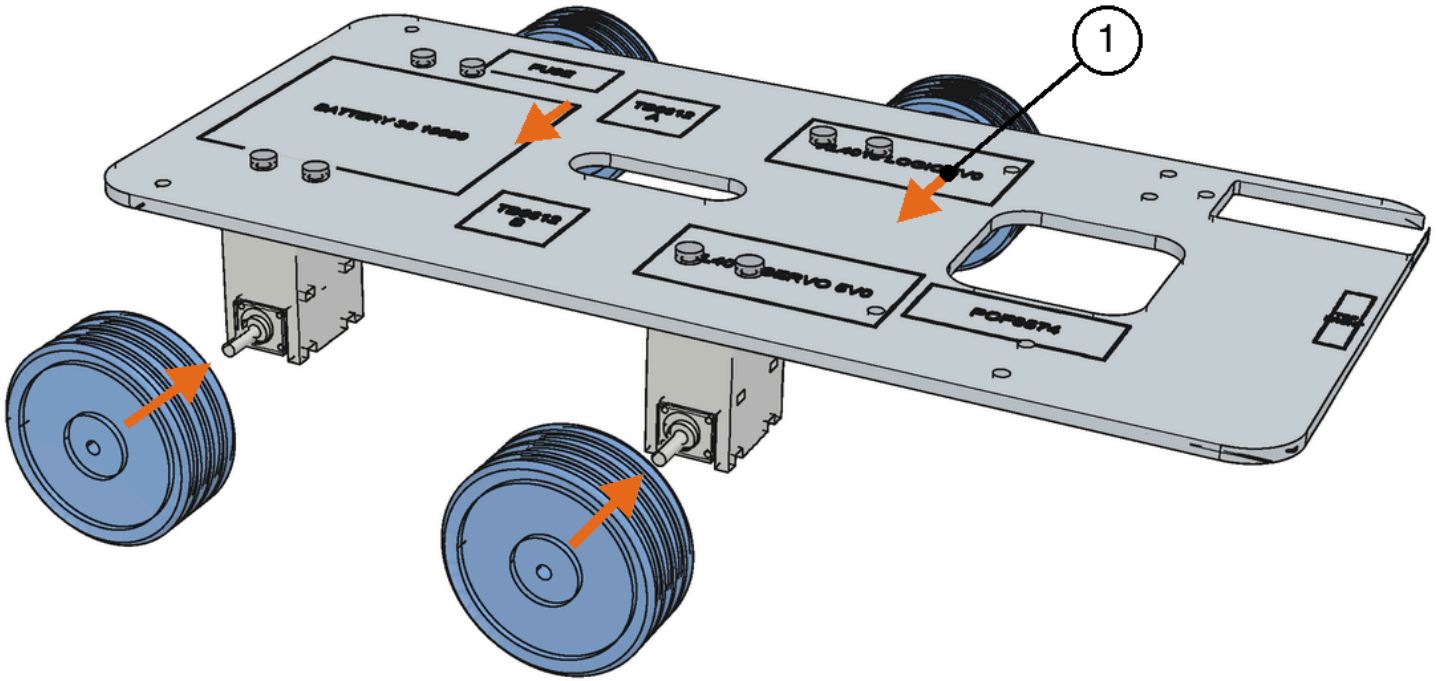
**RESULT**



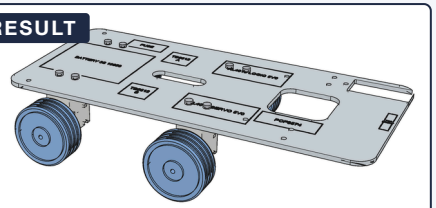
## YOU NEED

| | | |
|---|---|---|
| 1 |  | Lower deck (laser-cut) **x 1** item #16 |
| 2 |  | Motor + mount **x 4** steps 1-2 |
| 3 |  | M3 x 6 screw **x 8** item #15 |

# 4 Wheels




RESULT

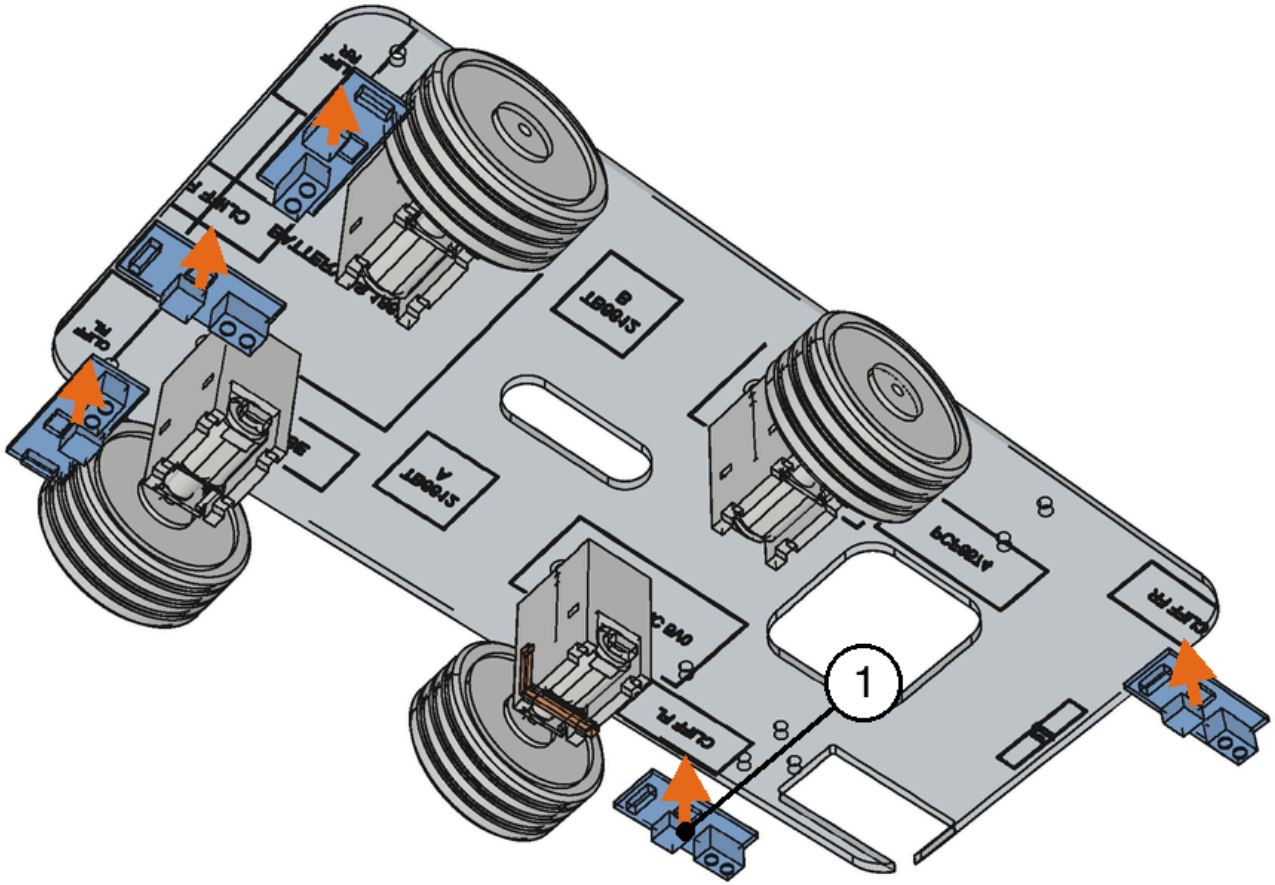


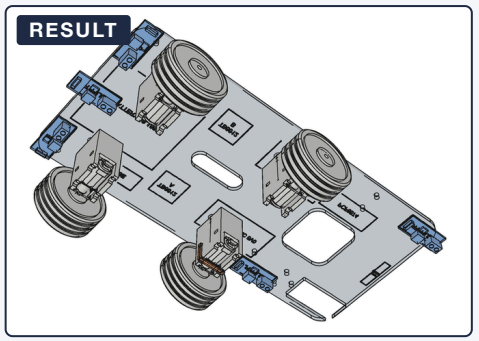
## YOU NEED

| | | |
|---|---|---|
| 1 |  | 44 mm wheel **x 4**<br>item #13 |

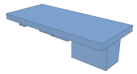
1. Line up the flat in the hub with the flat on the motor's D-shaft.
2. Press the wheel on until it stops. Spin each wheel by hand: nothing may rub.

# 5 Table-edge sensors (deck upside down)



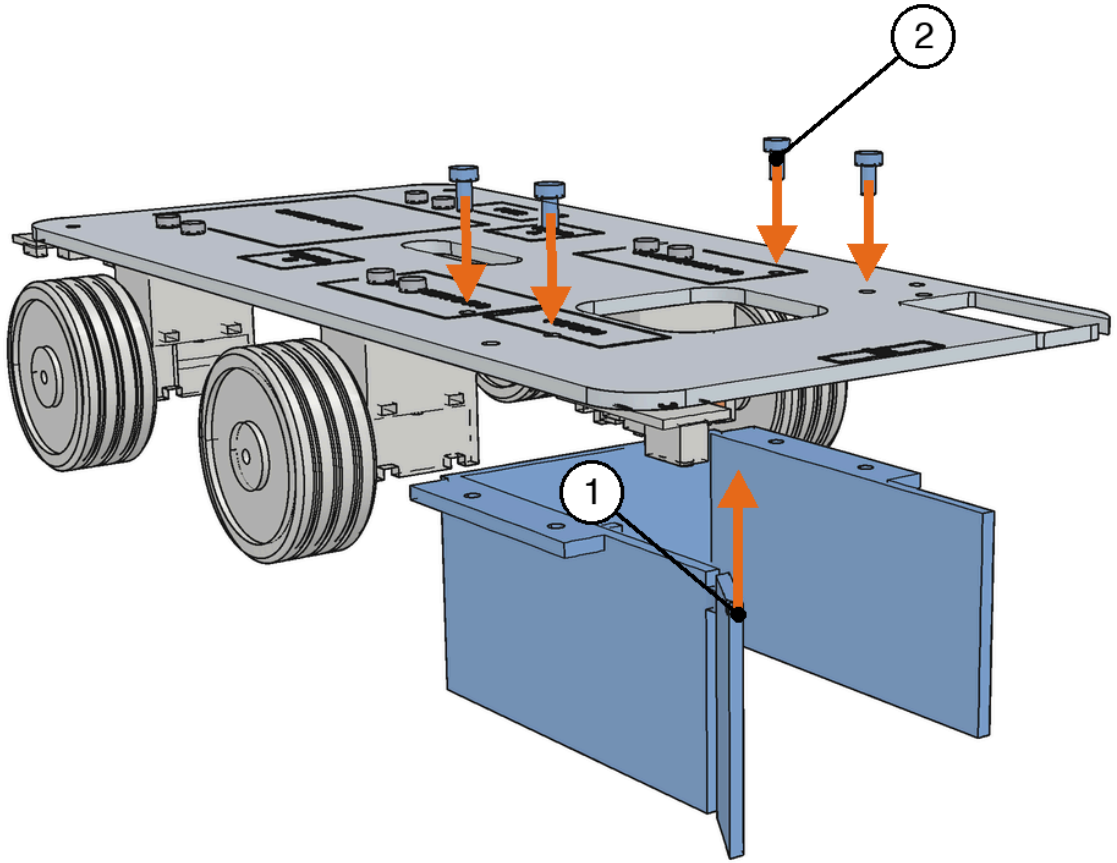

## YOU NEED

- (1)  TCRT5000 edge sensor **x 5** — item #18
- (·) Foam tape + paper template **x 1** — item #38

1. Turn the deck over. Tape the printed underside template on and stick each sensor on its outline,
2. lens down: front-left, front-right, rear, and the two rear corners.
3. Lens 3-8 mm above the table. Route each lead up through the harness slot.

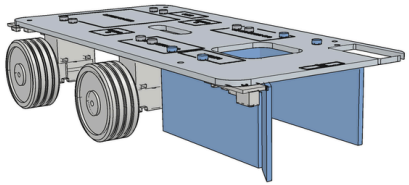
**Important.** These sensors stop the rover at the table edge. Every one is checked in bring-up before any run.

# 6 Intake shroud

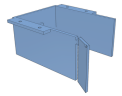



1. Hold the shroud under the front of the deck, flanges flat against it, open side forward.
2. Drive an M3 x 6 down through the deck into each flange's pilot hole; it cuts its own thread.

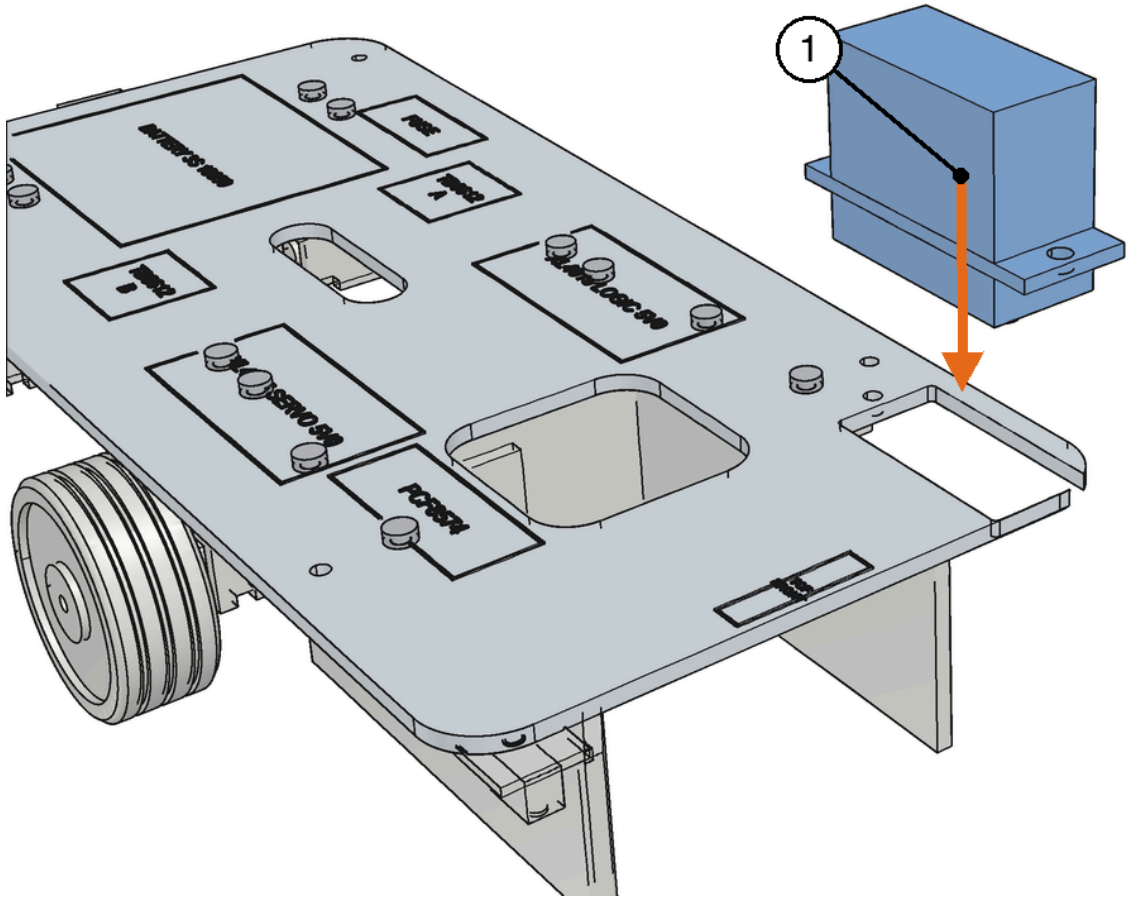
**RESULT**



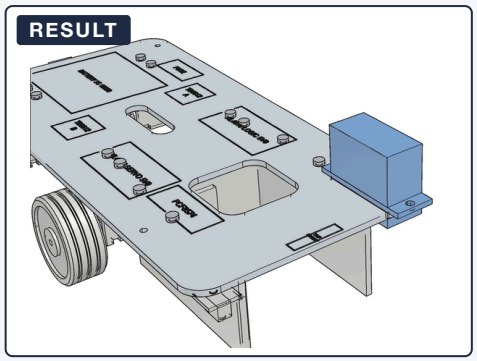
**YOU NEED**

| | | |
|---|---|---|
| 1 |  | Intake shroud (printed) **x 1**<br>PLA, plate 1 |
| 2 |  | M3 x 6 screw **x 4**<br>item #15 |

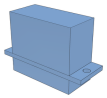
# 7 Door servo



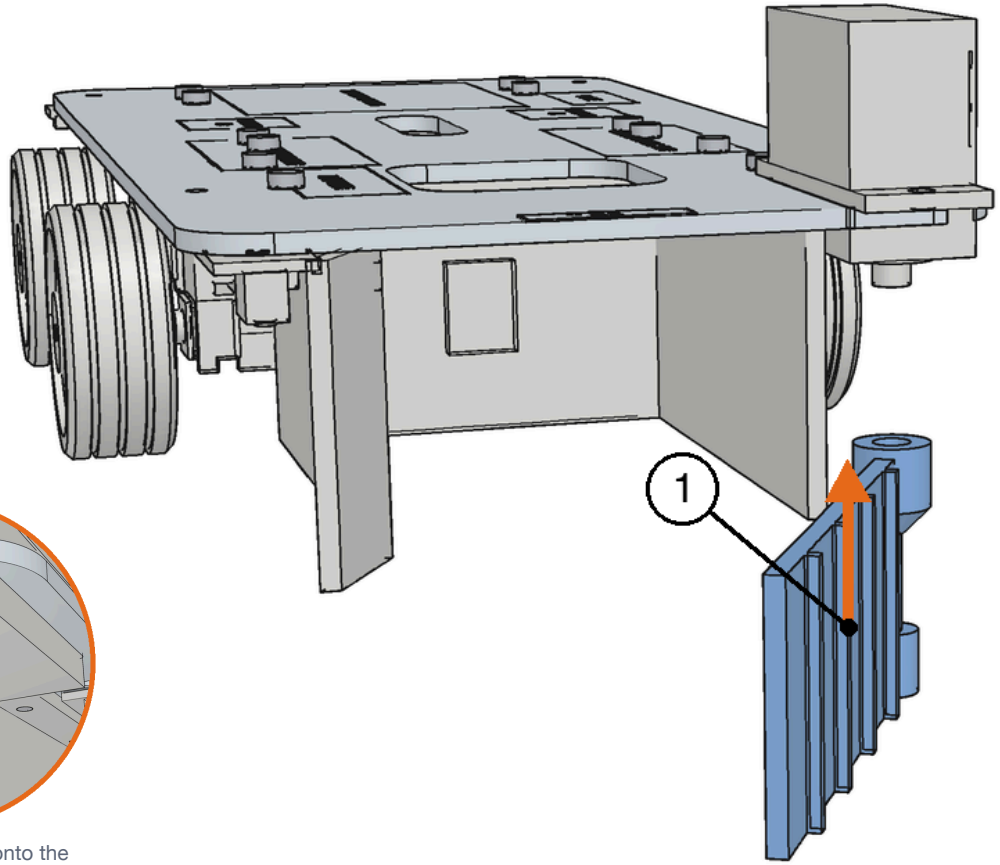

1. Lower the servo, output spline DOWN, into the rectangular pocket at the front-left of the deck.
2. Its ears rest on the deck. Tie each ear down with a cable tie through the ear hole and the deck hole.
3. Power it once from bring-up and send it to CLOSED before fitting the door.

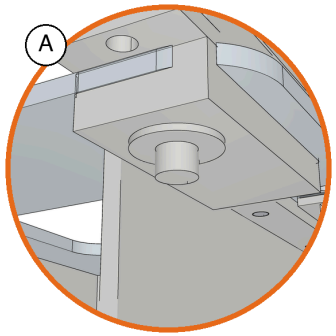



## YOU NEED

| | | |
|---|---|---|
| 1 |  | MG946R servo **x 1**<br>item #9 |
| • | | Cable tie, 100 mm **x 2**<br>item #36 |

# 8 Intake door

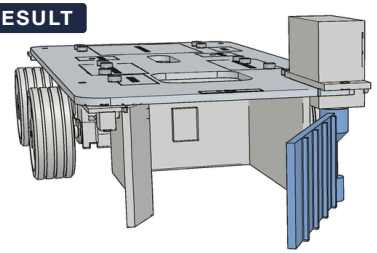



A: door hub pressed onto the spline

1. Warm the servo spline for a few seconds with a hairdryer.
2. Press the door's hub straight up onto the spline, door across the bay mouth (CLOSED).
3. If it ever slips, glue the servo's round horn to the hub instead.

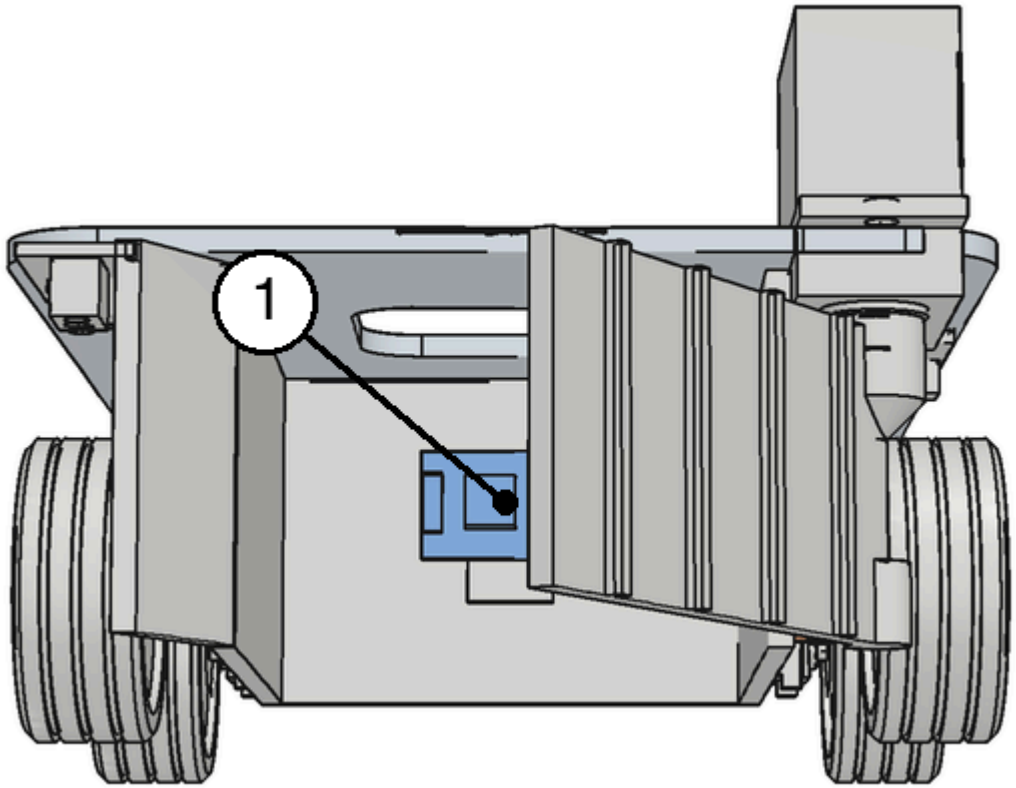
RESULT



## YOU NEED

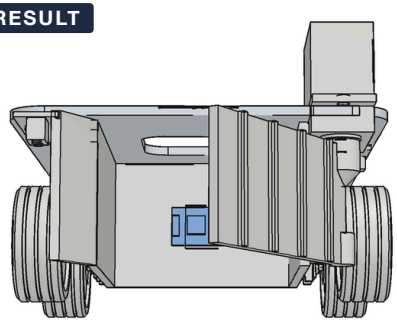
| | | |
|---|---|---|
| 1 |  | Intake door (printed) x 1<br>PLA, plate 1 |

# 9 Bay sensor

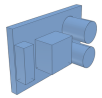



1. Reach in through the bay mouth (door open) and stick the sensor on the boss on the back wall,
2. lens facing forward, out of the bay.
3. It must see the token only once the token is fully inside.

RESULT

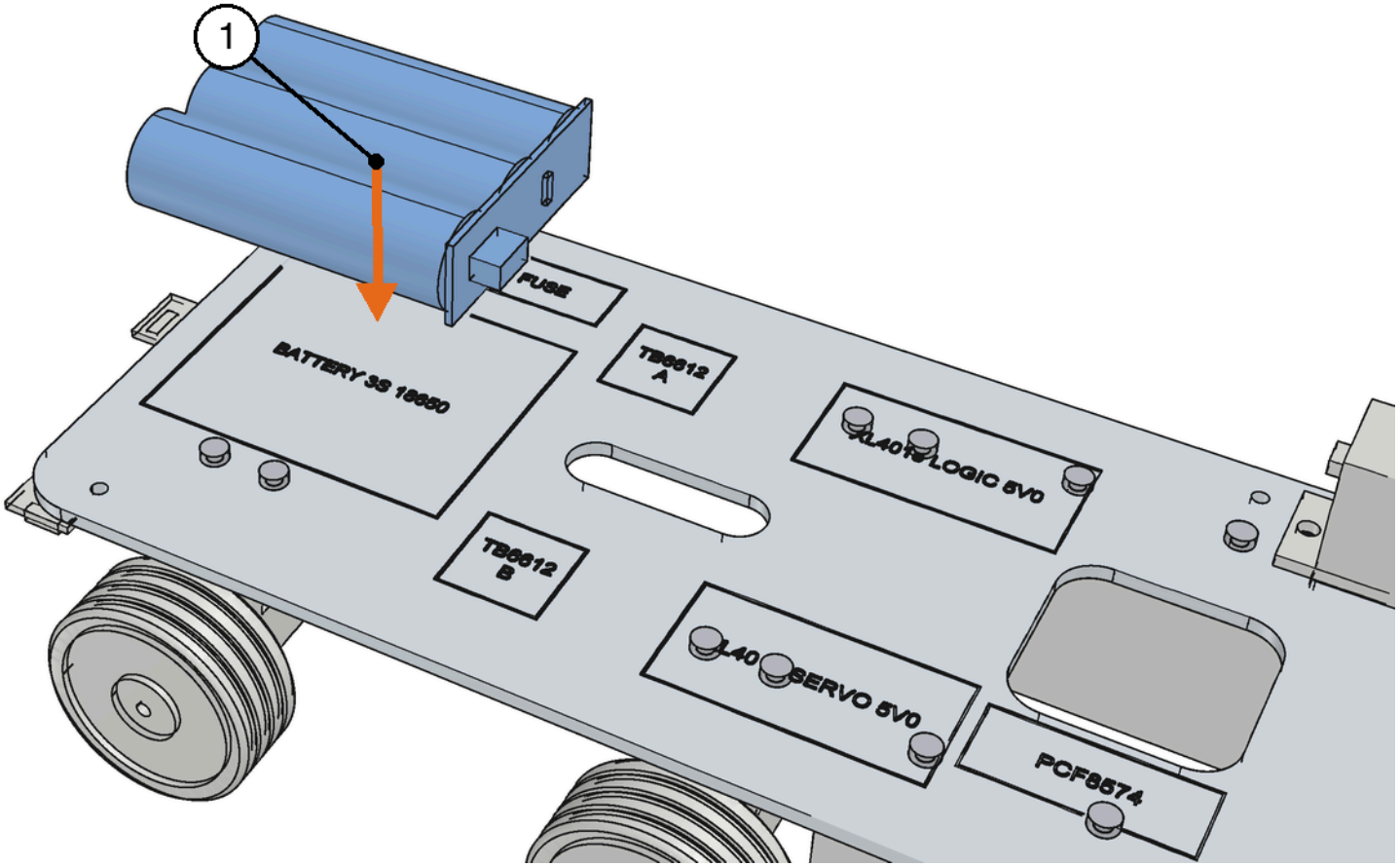


## YOU NEED

1  IR obstacle sensor **x 1**
item #19

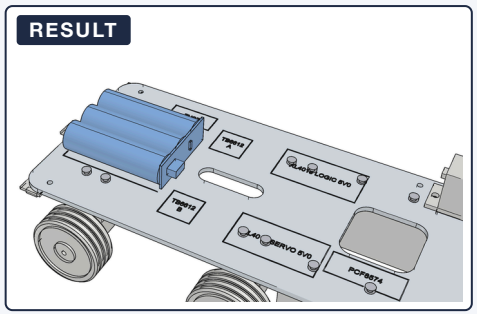
• Foam tape **x 1**
item #38

# 10 Battery

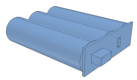



1. Place the pack on its outline at the rear of the deck, leads toward the middle.
2. Strap it down with two cable ties through the deck. Leave it UNPLUGGED.

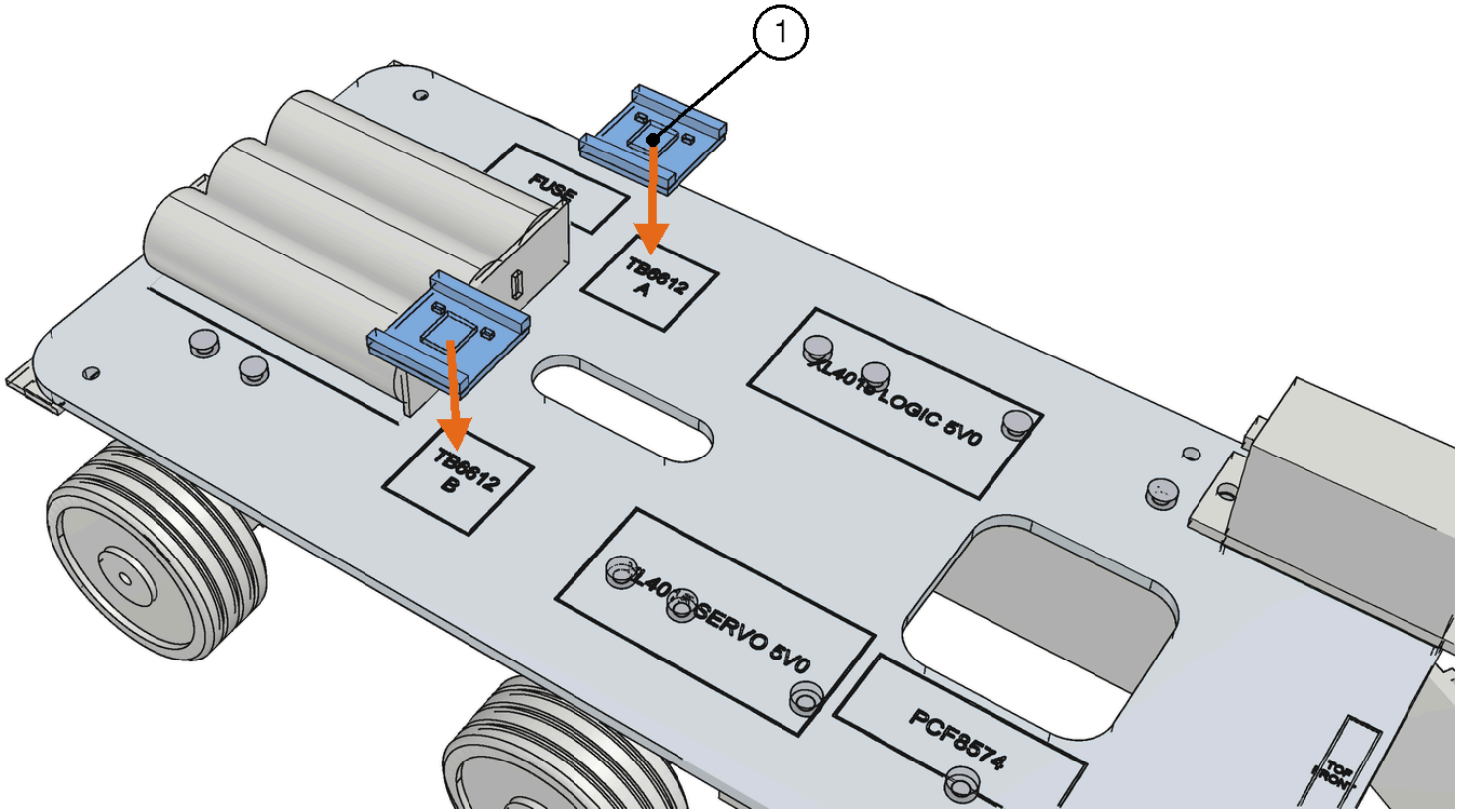
**Important.** The battery stays unplugged until both regulators are set to 5.0 V (step 12).




## YOU NEED

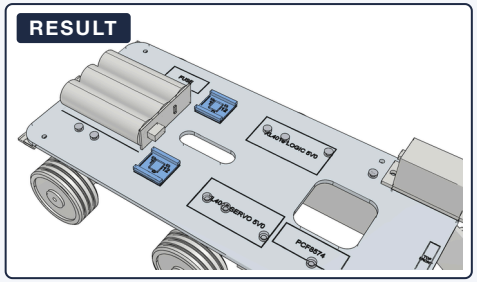
- (1)  3S 18650 pack **x 1** — item #10
- (·) Cable tie, 100 mm **x 2** — item #36

# 11 Motor drivers

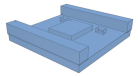



1. Stick each driver on its outline (A left, B right), pins up.

RESULT

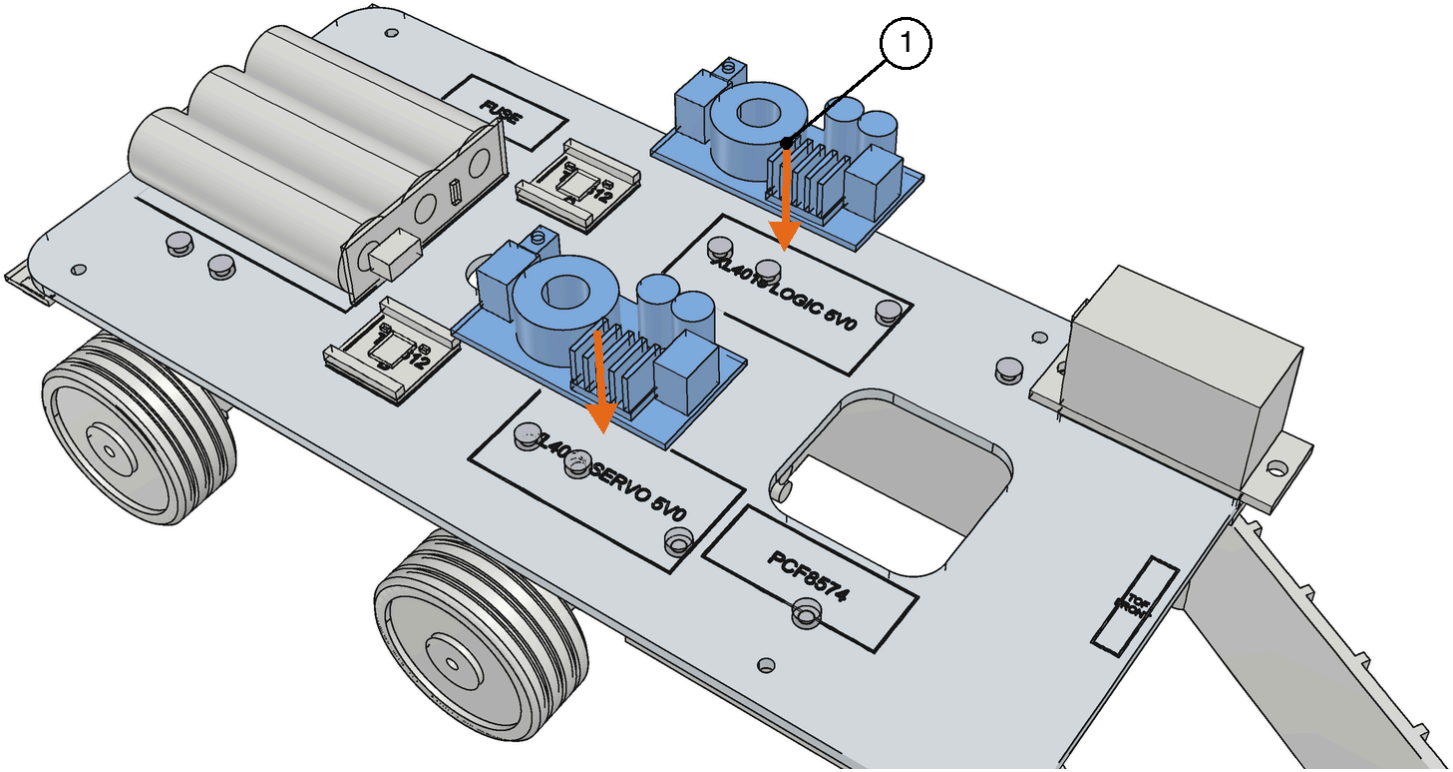


YOU NEED

1  TB6612FNG driver **x 2**
item #6

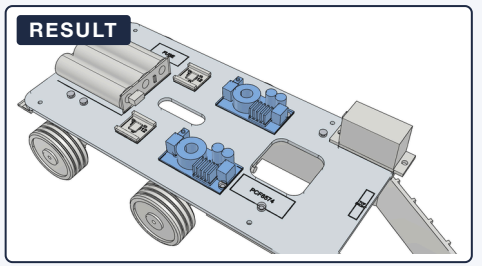
• Foam tape **x 1**
item #38

# 12 Voltage regulators

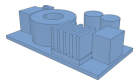



1. Three screw heads sit under each regulator's outline. Build two 3-layer foam-tape pads
2. per regulator, beside the heads, and stick the board on them: LOGIC left, SERVO right.
3. Feed each input from the battery, turn its trimmer until a multimeter reads 5.0 V at the
4. output, then unplug the battery again.

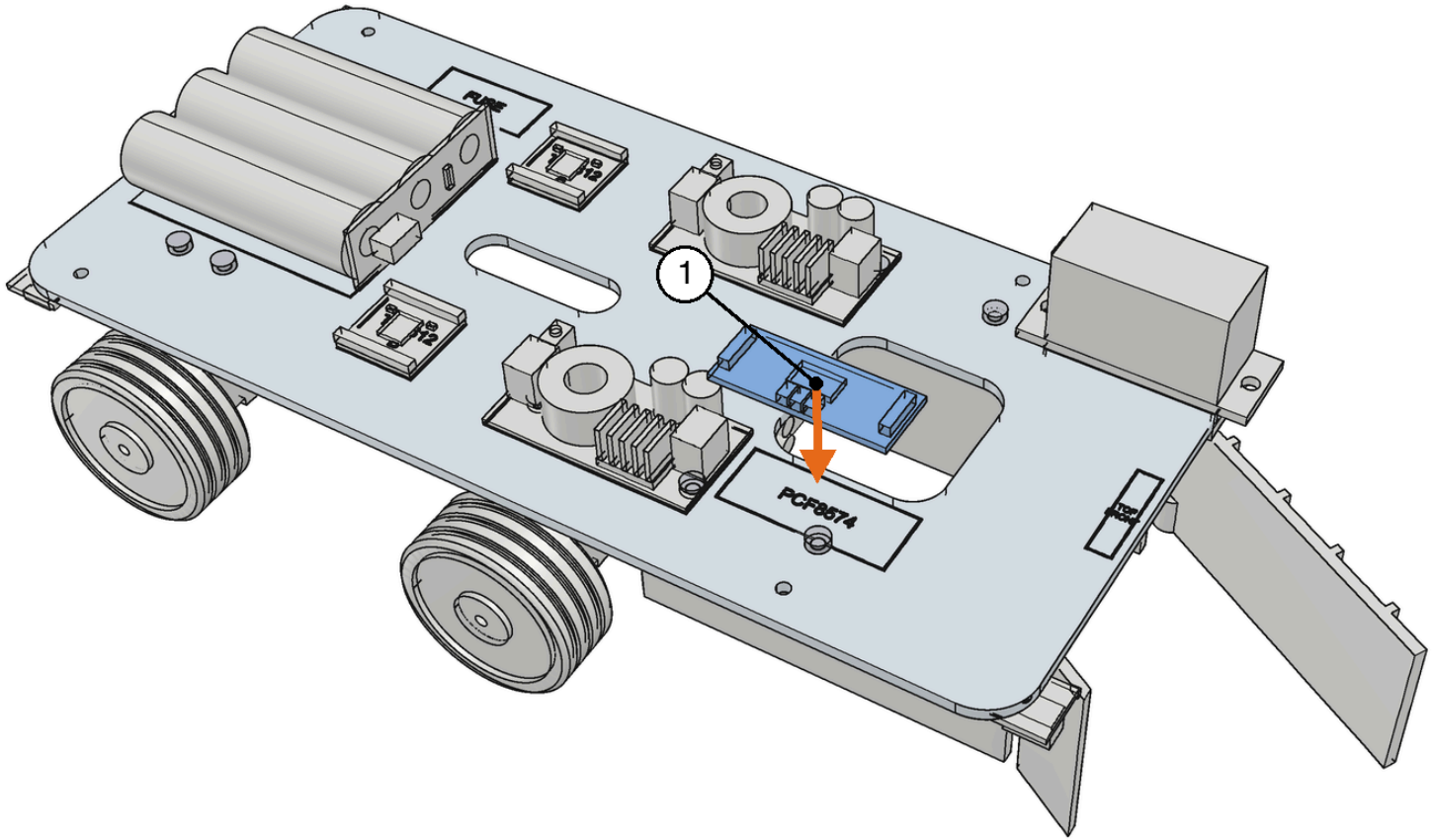
**Important.** 5.0 V first. At its factory setting a regulator can put out 12 V and destroy the ESP32 and the servo.




## YOU NEED

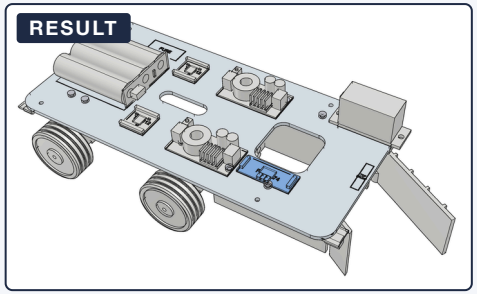
- 1 —  XL4015 regulator **x 2** — item #12
- • Foam tape, 3 layers **x 1** — item #38

# 13 I/O expander

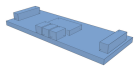



1. A shroud screw head sits under one corner of this outline: build a 3-layer foam pad beside it.
2. Stick the PCF8574 on its outline, pins toward the middle of the deck.

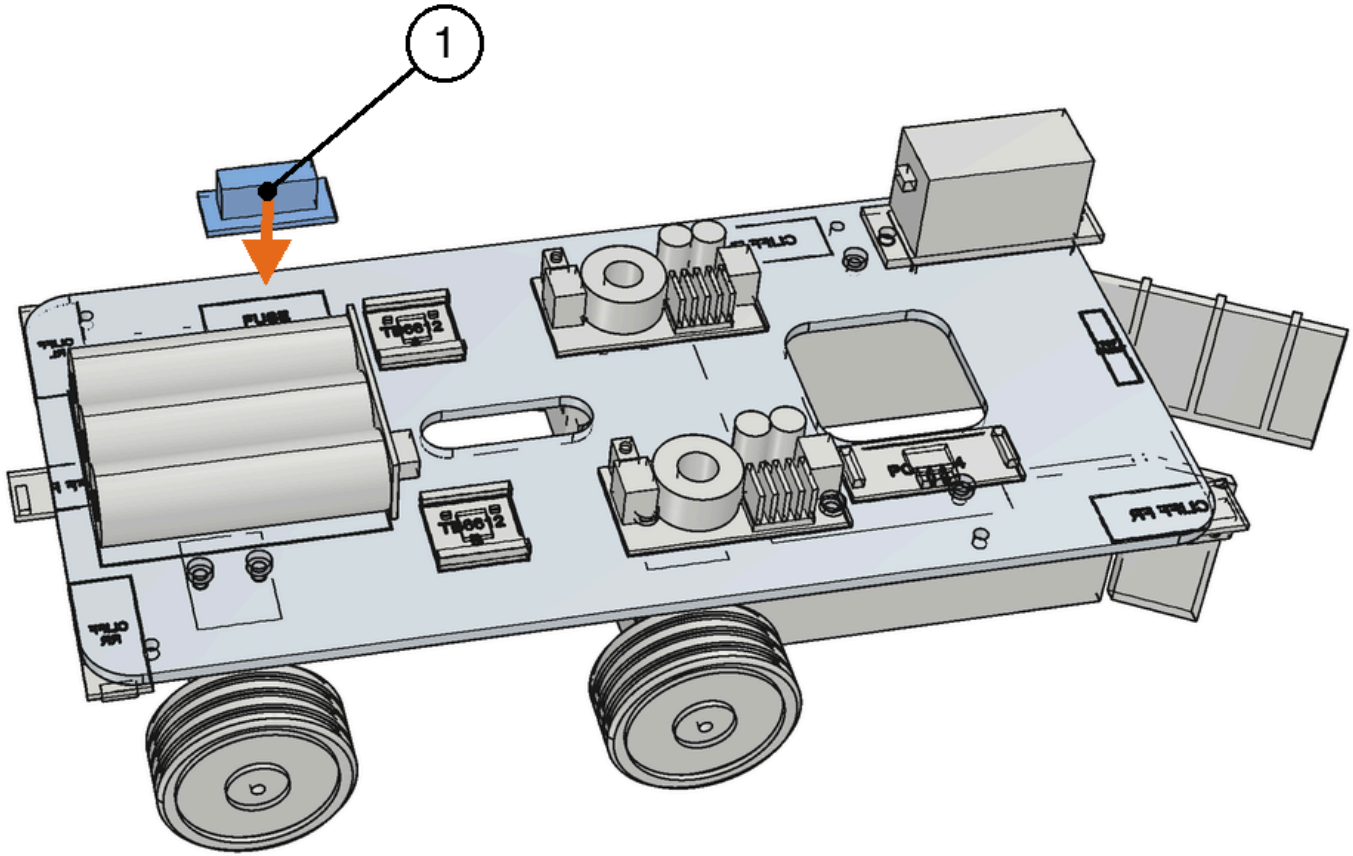
RESULT



## YOU NEED

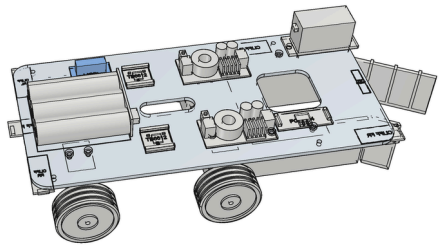
- (1)  PCF8574 I/O expander **x 1** — item #17
- (·) Foam tape, 3 layers **x 1** — item #38

# 14 Fuse board

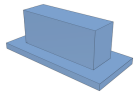



1. Push the 6.3 A fuse into its holder.
2. Stick the fuse board on its outline at the rear left, next to the battery.

**RESULT**

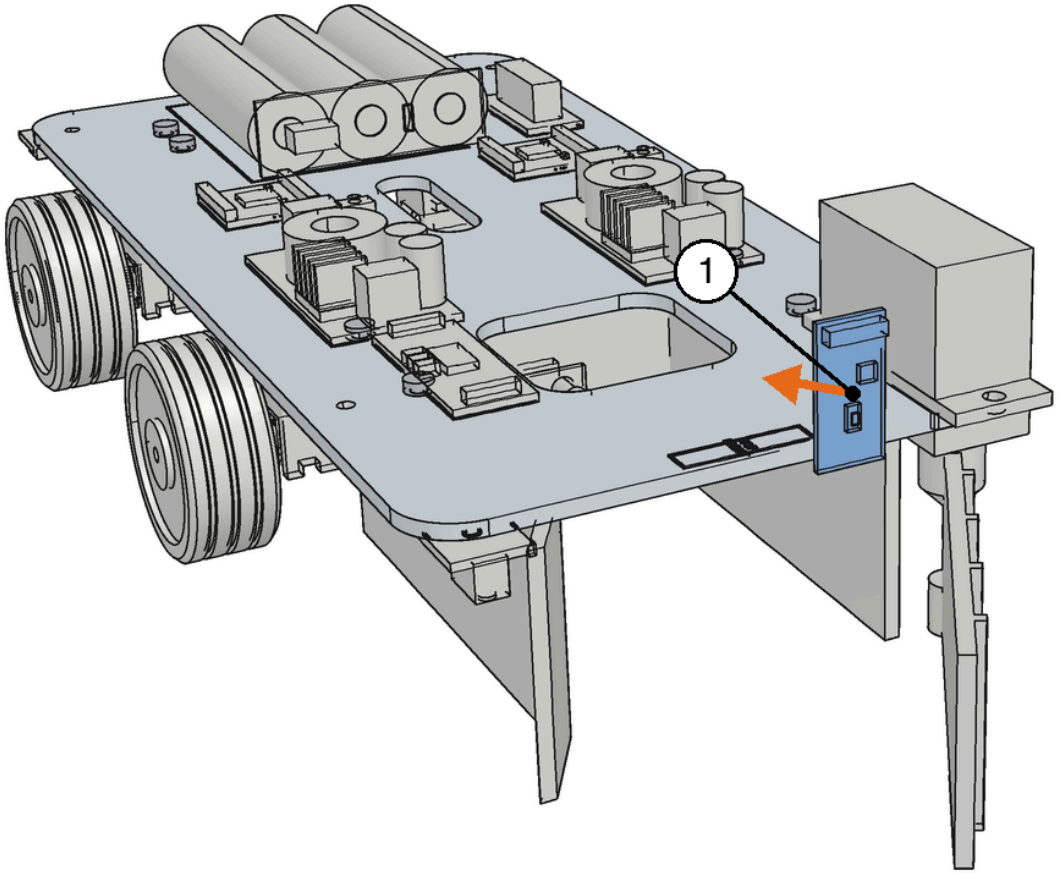


**YOU NEED**

1  Fuse board: holder + 6.3 A fuse **x 1**
item #22, #23

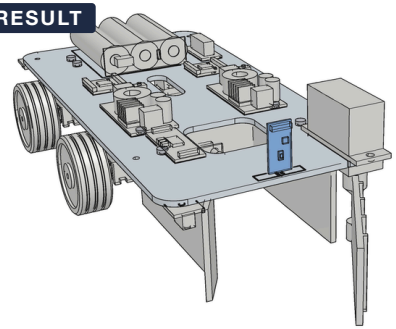
• Foam tape **x 1**
item #38

# 15 Front distance sensor




1. Tape the sensor to the upright face of an L-bracket, sensor window facing forward.
2. Tape the bracket to the front lip of the deck, centred, sensor looking straight ahead.

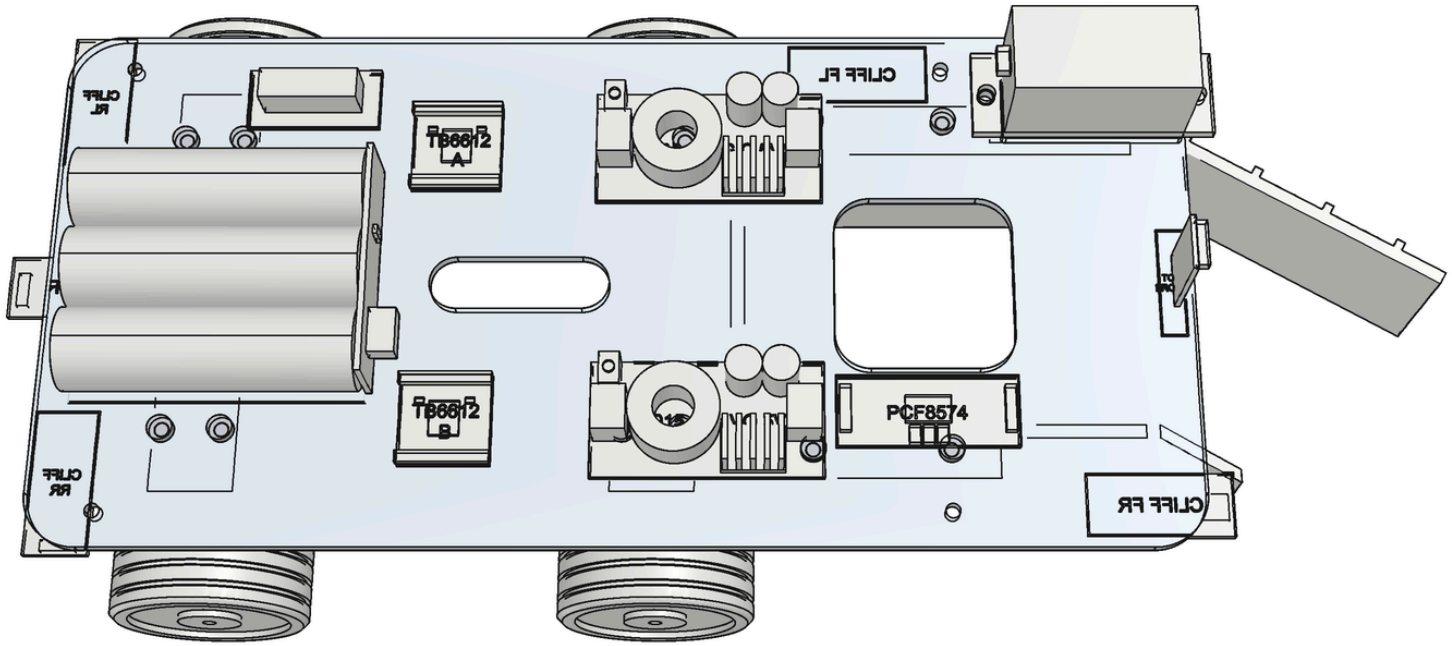
RESULT



## YOU NEED

- (1)  VL53L0X + L-bracket **x 1** — item #7, plate 1
- (·) Foam tape **x 1** — item #38

# 16 Wire the lower deck



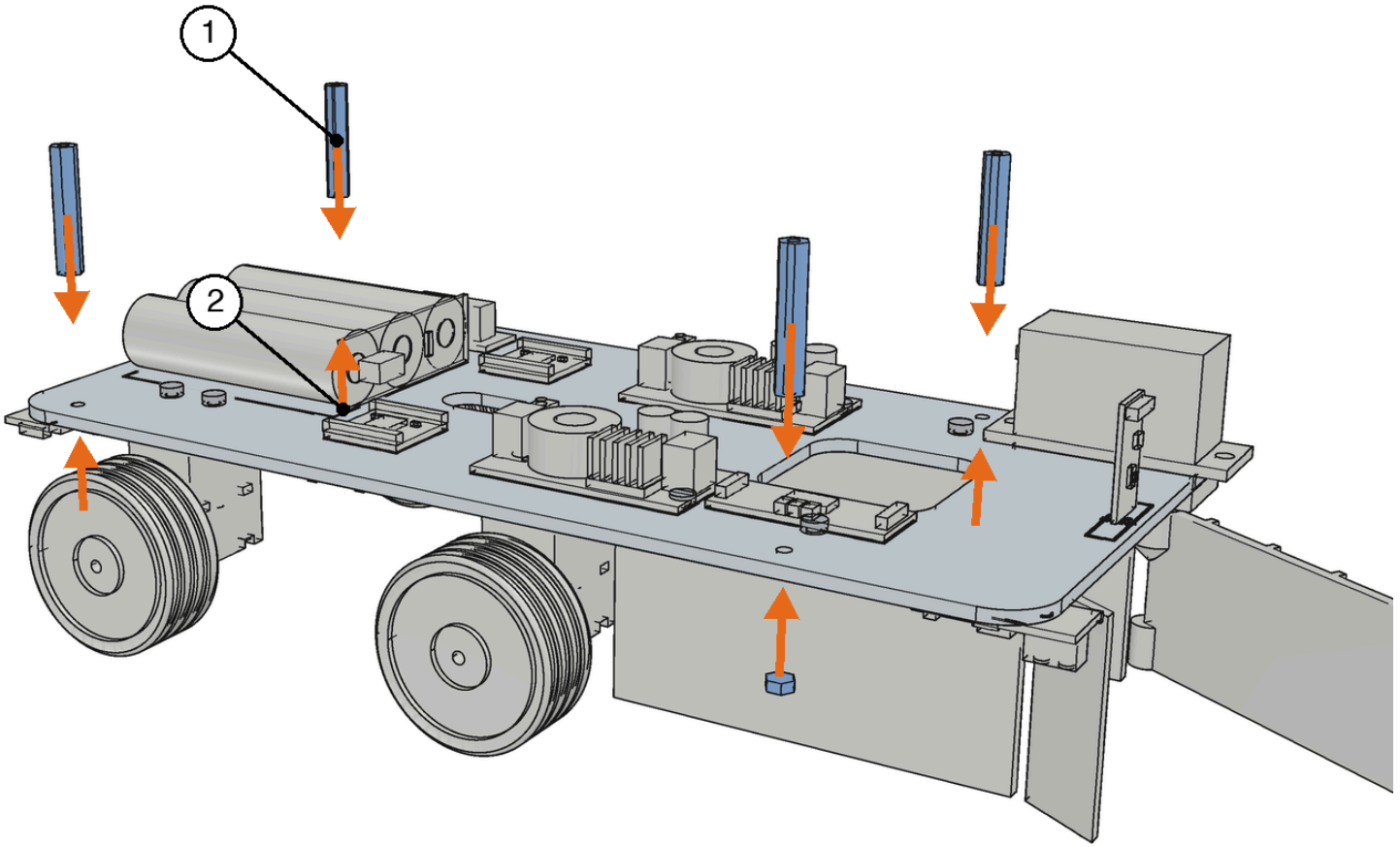

1. Open AEGIS_Wiring_Guide.pdf at W4 and do W4 to W9 in order, one table row at a time.
2. W4 power, W5 set both regulators to 5.0 V, W6 motors, W7 servo power, W8 edge and bay sensors.
3. W9: label every lead that goes to the upper deck and leave it 15 cm long at the harness slot.
4. Do each step's green CHECK before the next one.

**Important.** Battery unplugged the whole time. Heat-shrink every power joint before you go on.

## YOU NEED

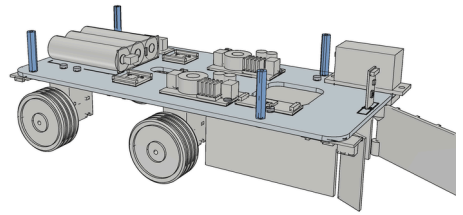
- Wiring guide, steps W4-W9 **x 1**
  item PDF
- Silicone wire, solder, heat-shrink **x 1**
  item #24-#35

# 17 Pillars

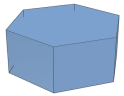



1. For each corner: push the stud of a 20 mm male-female pillar down through the hole and fit a nut
2. underneath. Screw a 10 mm male-female pillar into its top: 30 mm between the decks.

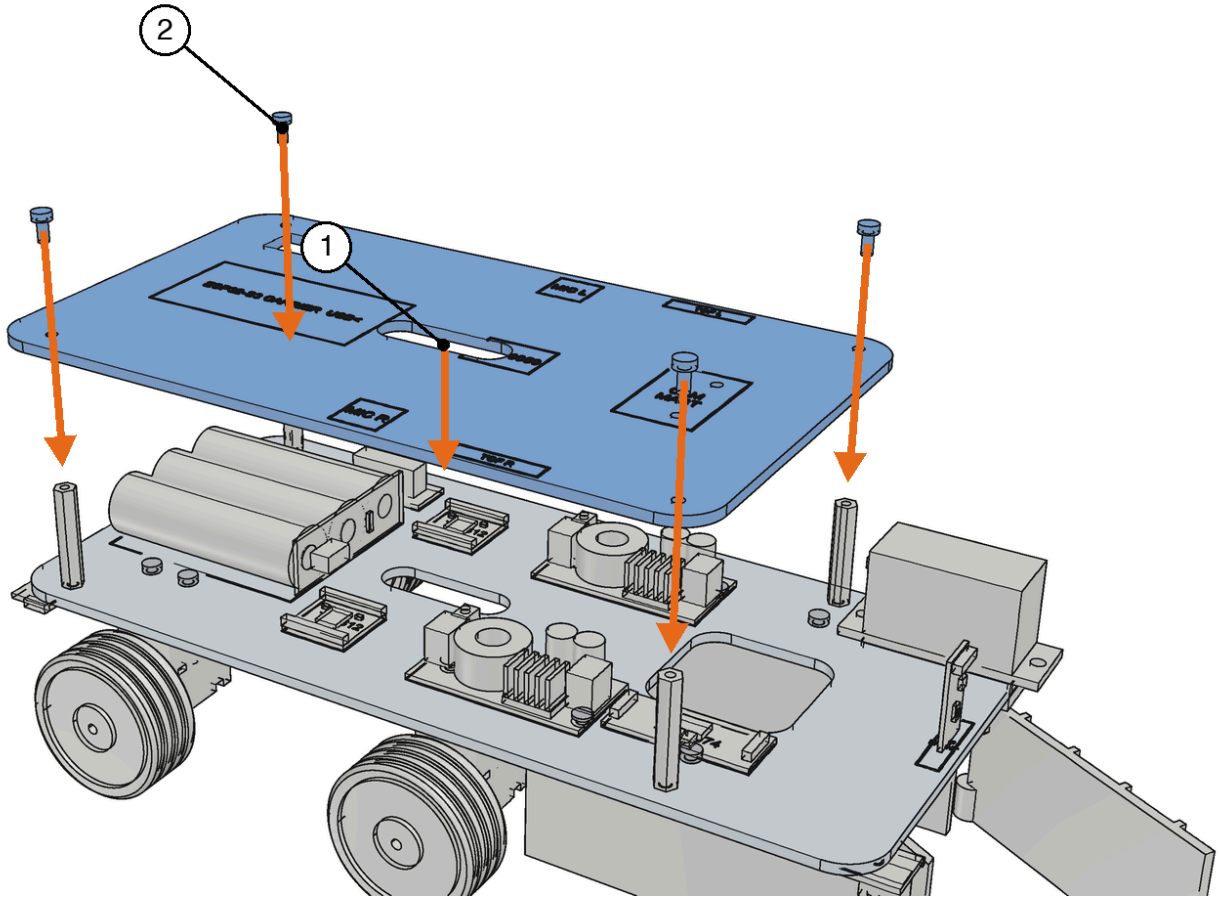
**RESULT**



## YOU NEED

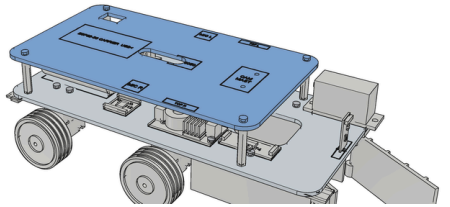
| | | |
|---|---|---|
| 1 |  | Pillar: 20 mm M-F + 10 mm M-F **x 4**<br>item #15 |
| 2 |  | M3 nut **x 4**<br>item #15 |

# 18 Upper deck

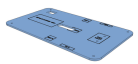



1. Feed the harness wires up through the slot in the upper deck.
2. Set the deck on the pillars, engraved side up, and fix it with four M3 x 6 screws.

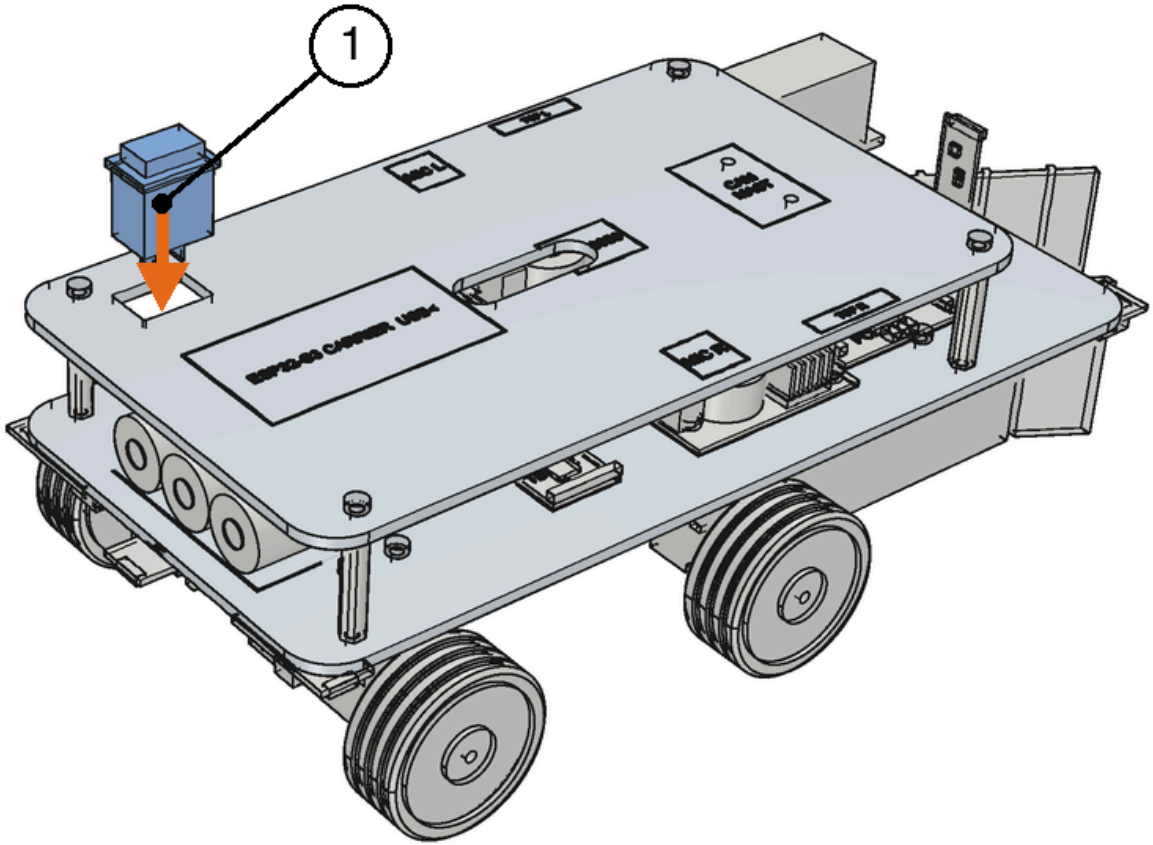
RESULT



YOU NEED

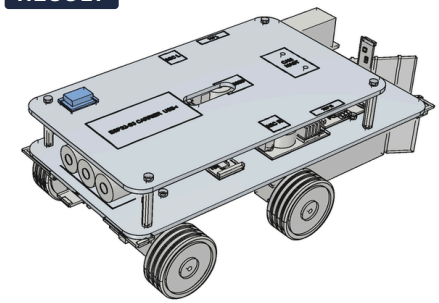
| | | |
|---|---|---|
| 1 |  | Upper deck (laser-cut) **x 1** item #16 |
| 2 |  | M3 x 6 screw **x 4** item #15 |

# 19 Power switch




1. Solder the battery + lead and the fuse-board lead to the switch first (wiring guide).
2. Push the switch down into its cut-out at the rear left until the clips snap. O = off.

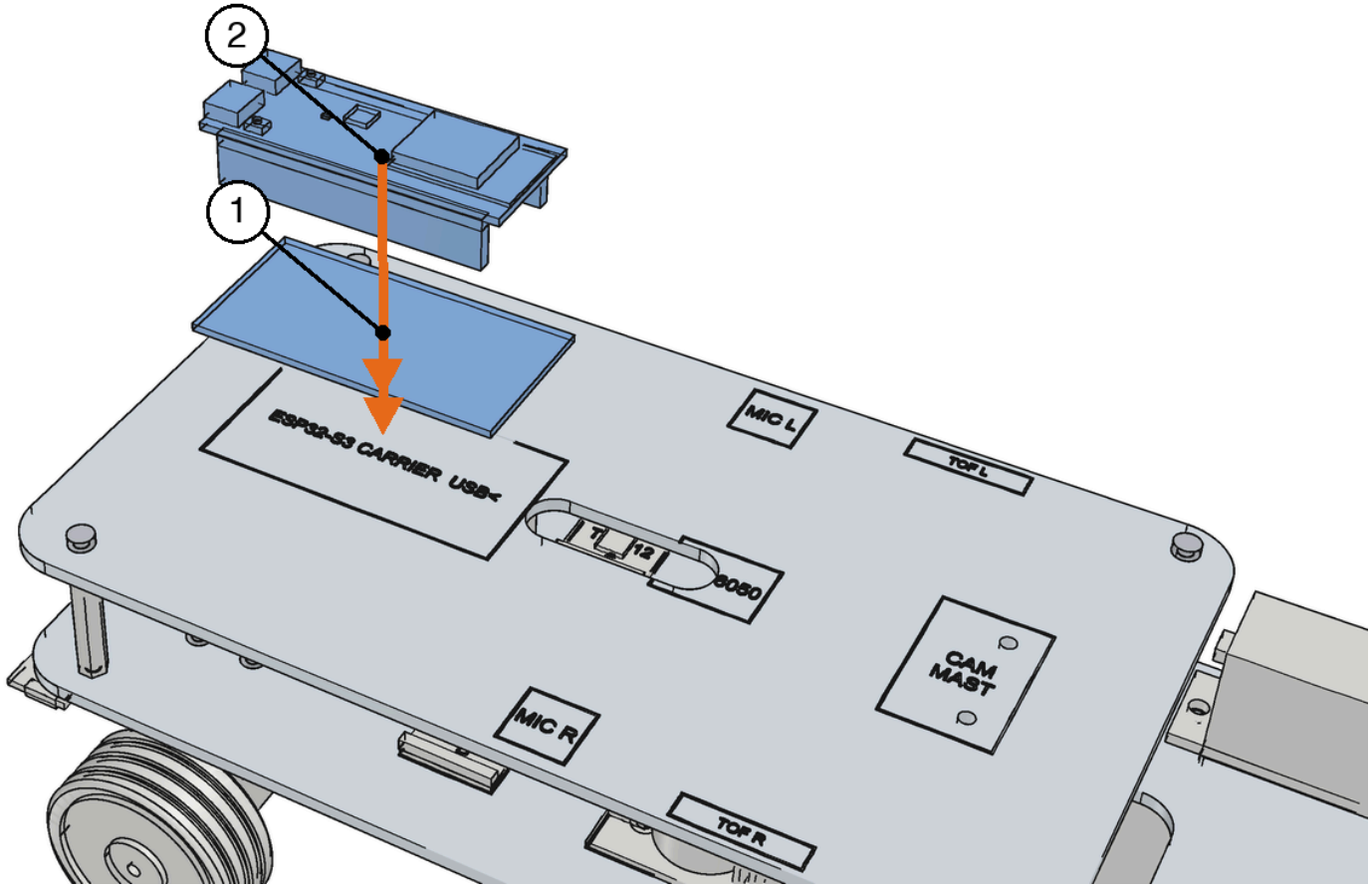
RESULT



## YOU NEED

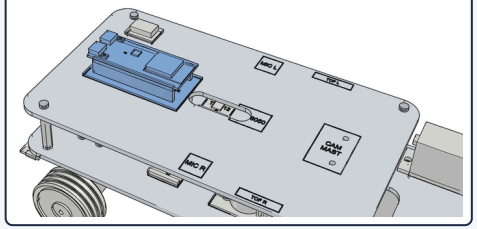
| | | |
|---|---|---|
| 1 |  | Rocker switch **x 1**<br>item #21 |

# 20 ESP32-S3 on its carrier board

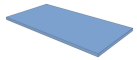


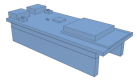

1. Stick the soldered carrier board on its outline with foam tape, sockets up.
2. Plug the ESP32-S3 into the sockets, USB ports toward the rear (the arrow on the deck).

**RESULT**

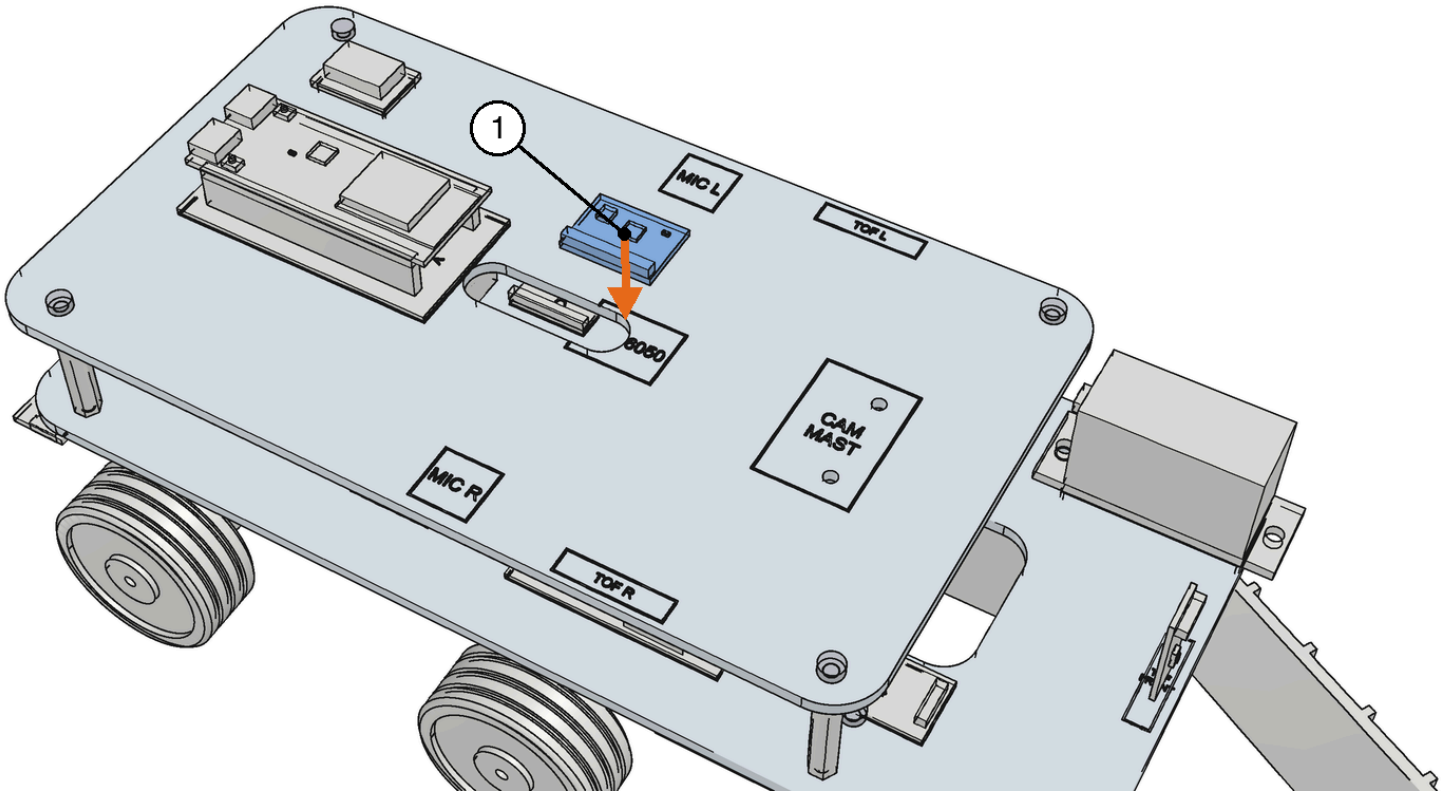


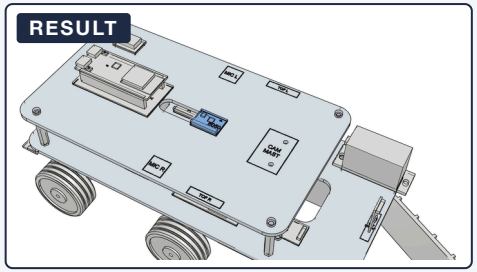
**YOU NEED**

1 
Carrier board, soldered **x 1**
item #31, #32

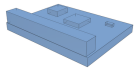
2 
ESP32-S3 DevKitC **x 1**
item #1

# 21 Gyro



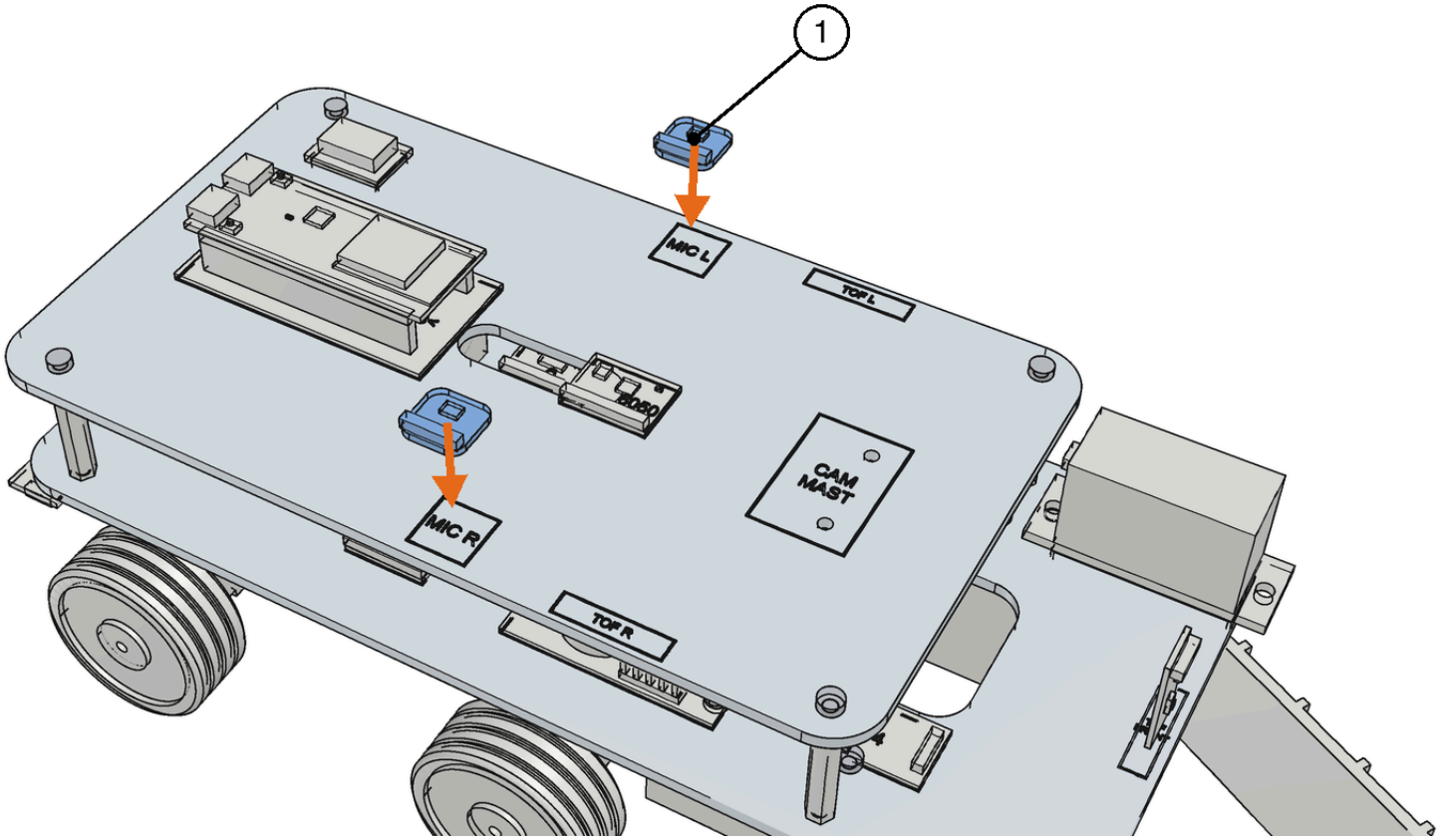

## YOU NEED

1.  MPU-6050 gyro **x 1**
   item #8

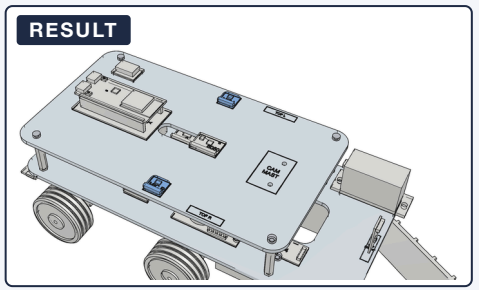
- Foam tape **x 1**
  item #38

1. Stick the MPU-6050 flat on its outline, edges square to the deck edge.
2. It measures the rover's heading: a tilted or skewed gyro makes the rover drive crooked.

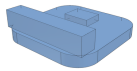
# 22 Microphones



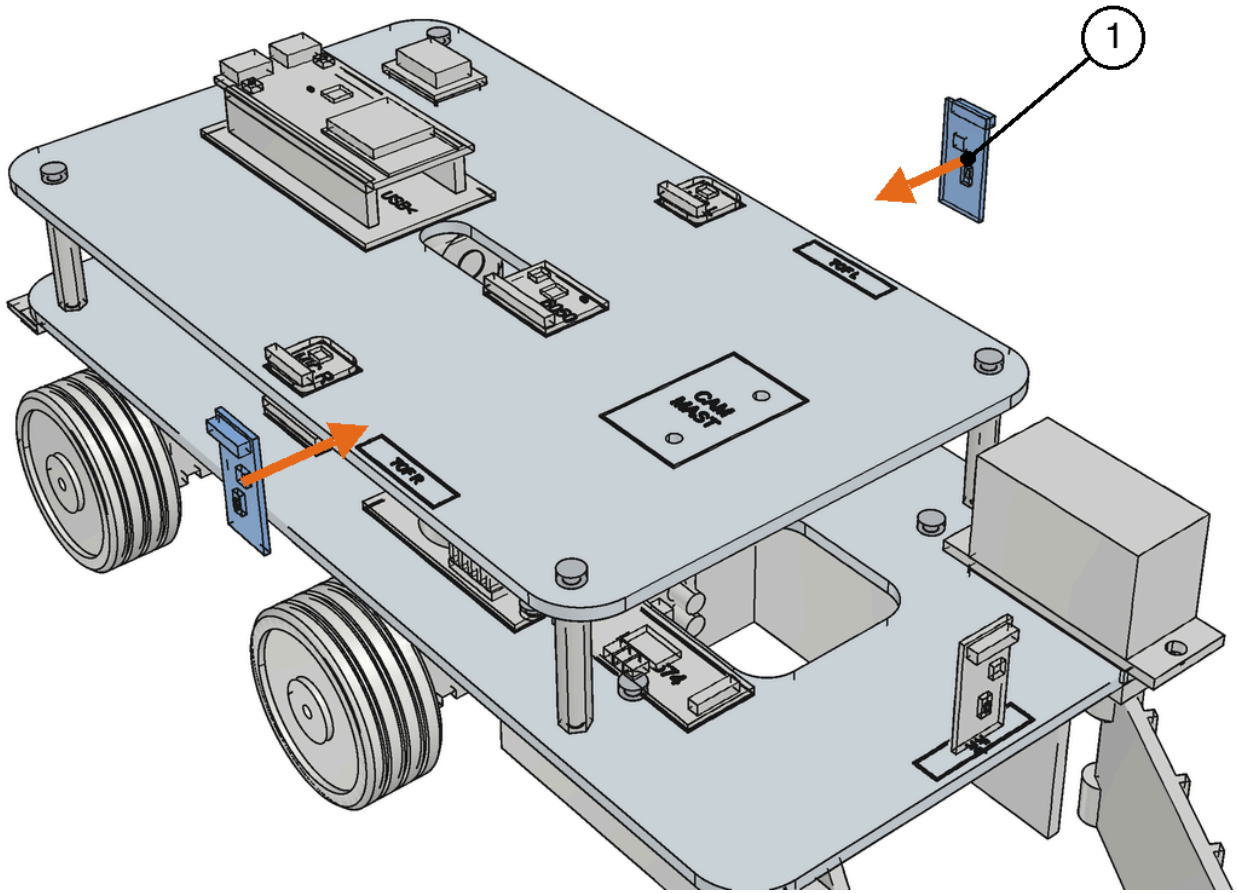

1. Stick one microphone on each outline, sound hole facing up: 100 mm apart, left and right.
2. Keep tape off the sound hole.



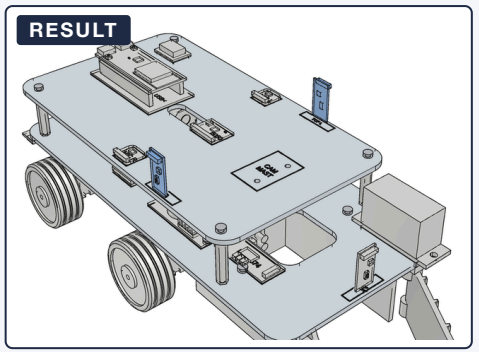

## YOU NEED

- (1)  INMP441 microphone **x 2** — item #4
- (·) Foam tape **x 1** — item #38

# 23 Side distance sensors



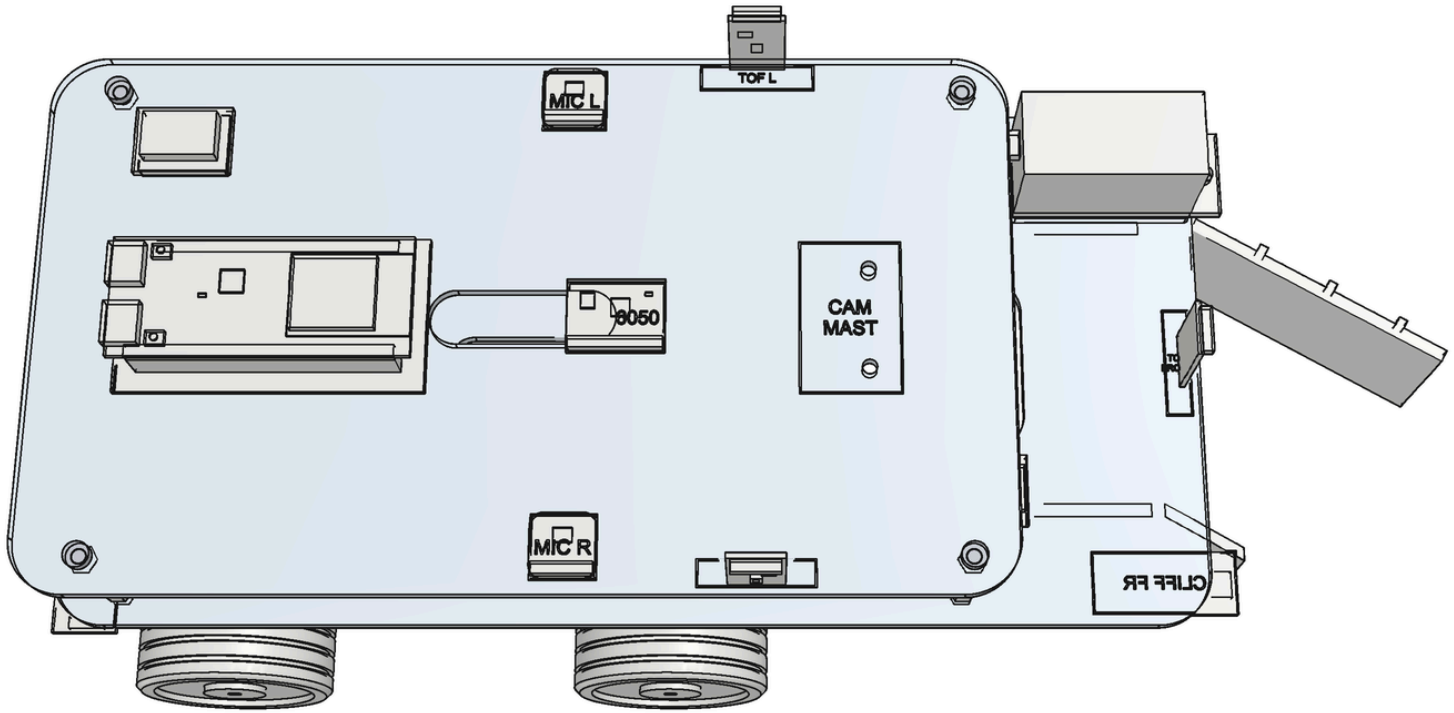

1. Tape each sensor to an L-bracket and the brackets on the deck edges, sensors facing out.
2. Point both straight out to the sides.




**YOU NEED**

| | | |
|---|---|---|
| 1 |  | VL53L0X + L-bracket<br>**x 2**<br>item #7, plate 1 |

# 24 Wire the upper deck

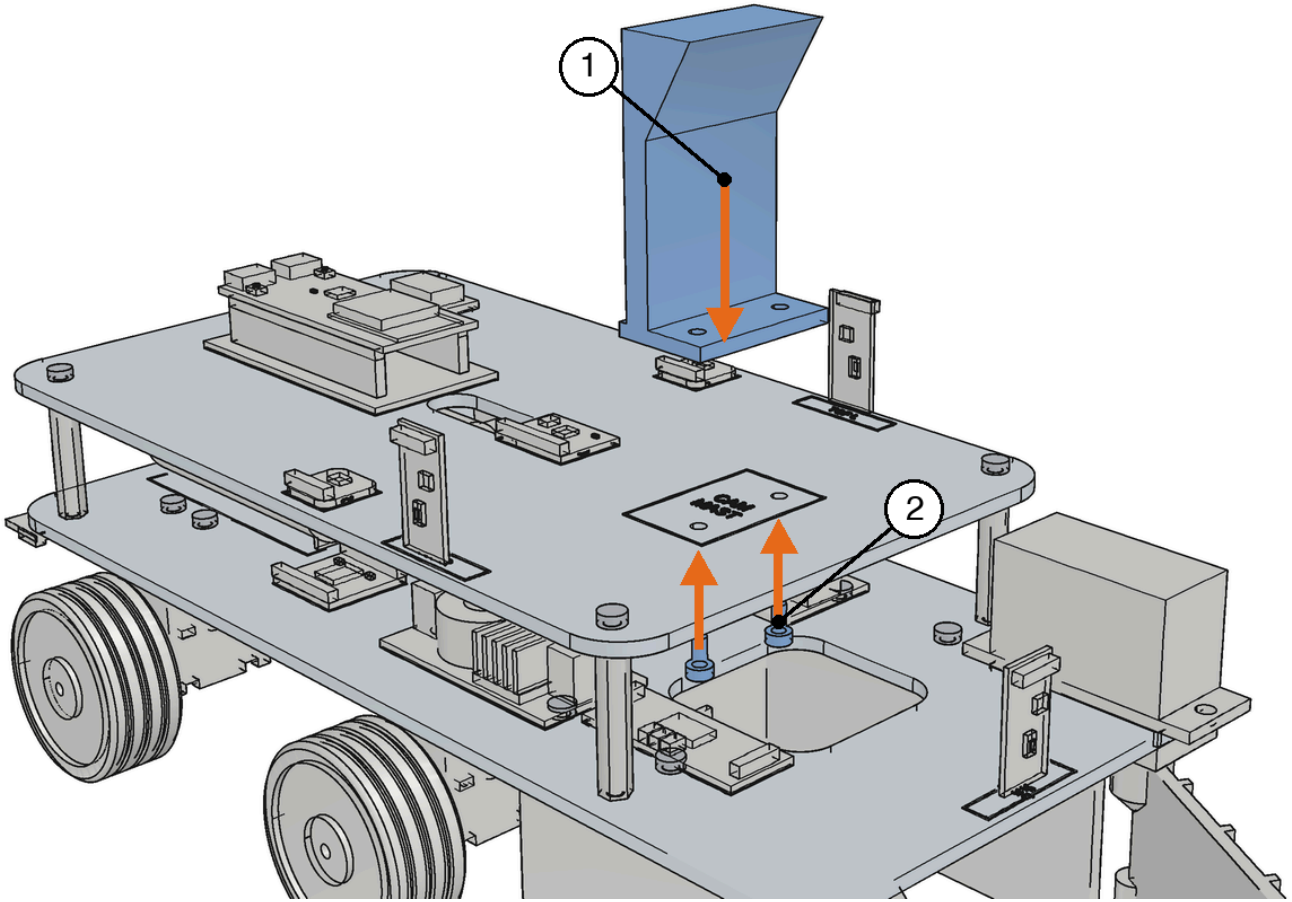



1. Take the ESP32-S3 out of its sockets. Then do W10 to W14 in the wiring guide, in order:
2. W10 power, W11 I2C sensors, W12 microphones, W13 motor control, W14 encoders and servo signal.
3. W15: first power-up checks, then plug the ESP32-S3 back in.
4. Bundle the wires with cable ties, clear of the wheels and the camera's view.

## YOU NEED

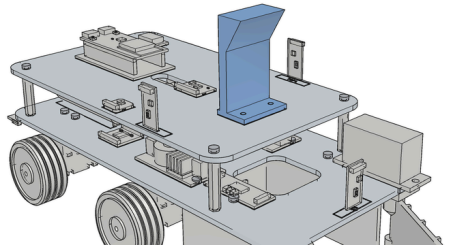
- Wiring guide, steps W10-W15 **x 1**
  item PDF

# 25 Camera mast




1. Stand the mast on its outline, sloped pad facing forward.
2. Drive two M3 x 6 screws up from under the upper deck into the pilot holes in its base.

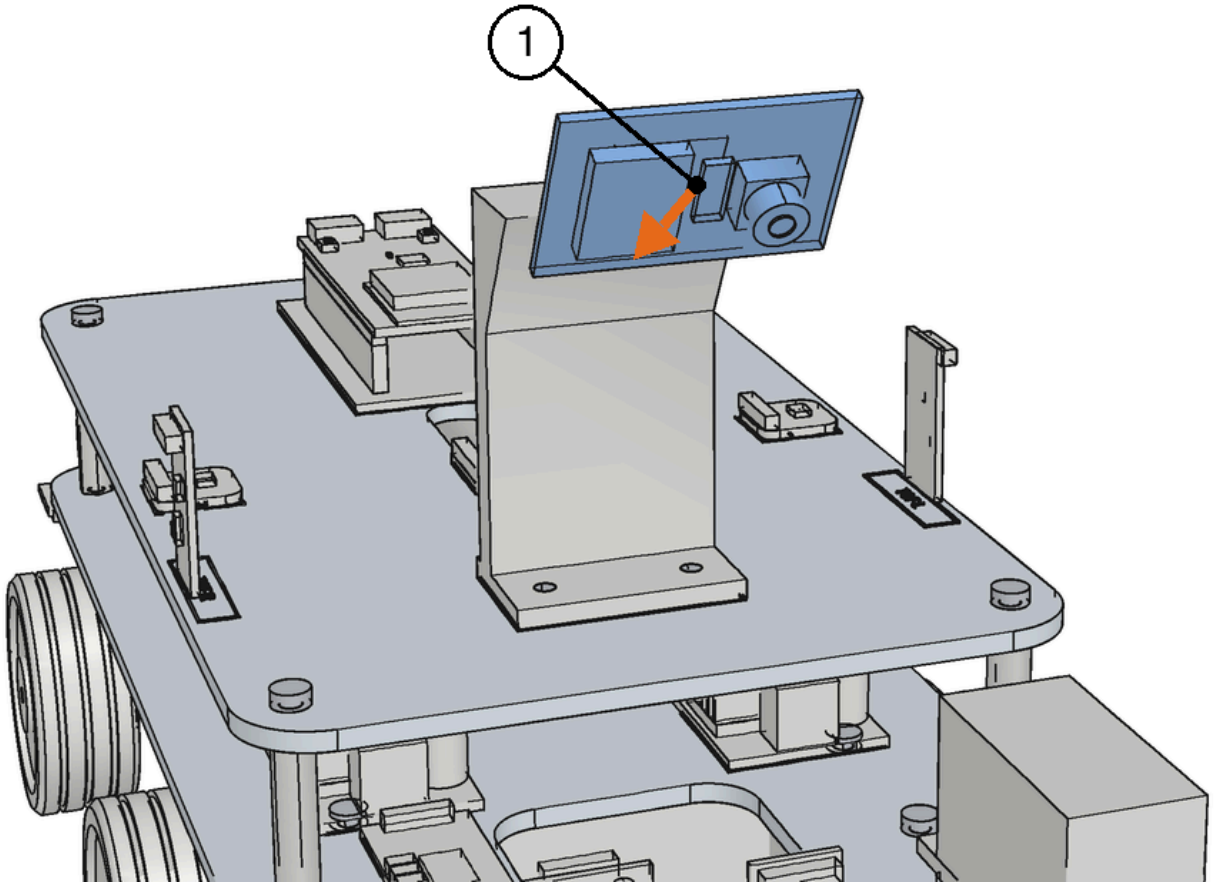
RESULT



## YOU NEED

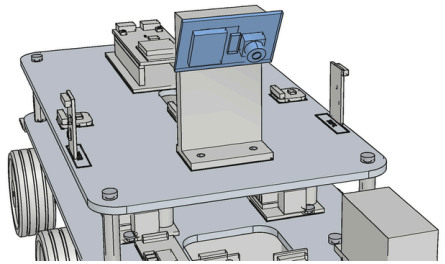
| | | |
|---|---|---|
| 1 |  | Camera mast (printed) **x 1** PLA, plate 1 |
| 2 |  | M3 x 6 screw **x 2** item #15 |

# 26 Camera

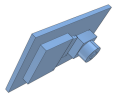



1. Put two layers of foam tape on the sloped pad, clear of the SD-card slot.
2. Press the ESP32-CAM on, lens forward: it looks 30 degrees down at the table.

RESULT

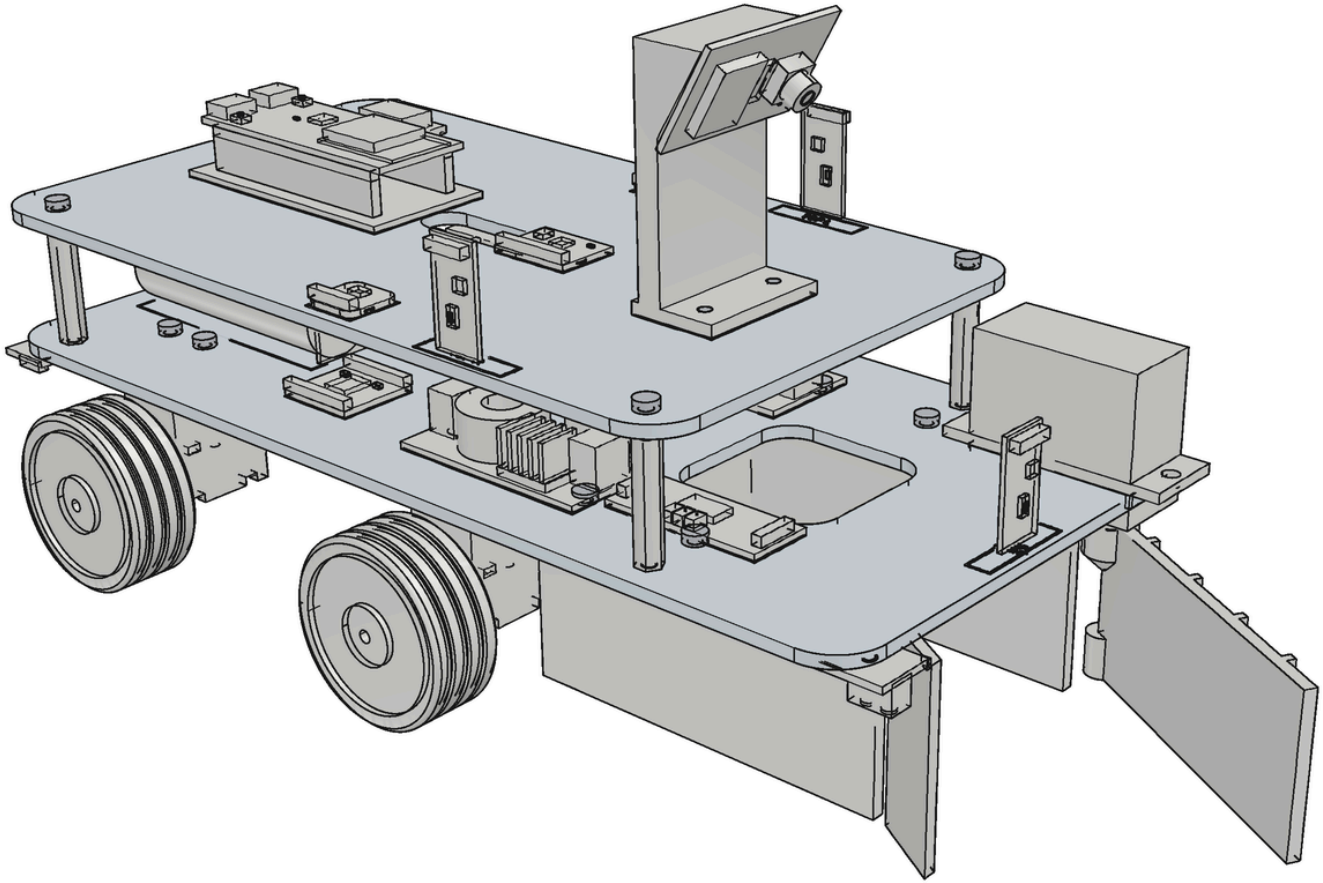


## YOU NEED

1  ESP32-CAM, flashed x 1
item #2

• Foam tape, 2 layers x 1
item #38

# 27 Finished: final checks



**YOU NEED**

Nothing new: checks only.

1. Every screw tight, no wire near a wheel, the door swings freely, the bay is clear.
2. Switch OFF, plug in the battery, switch ON.
3. Follow AEGIS_Deployment_Guide.pdf: flash both boards, bring-up checks, and the table-edge test
4. before the first autonomous run.